

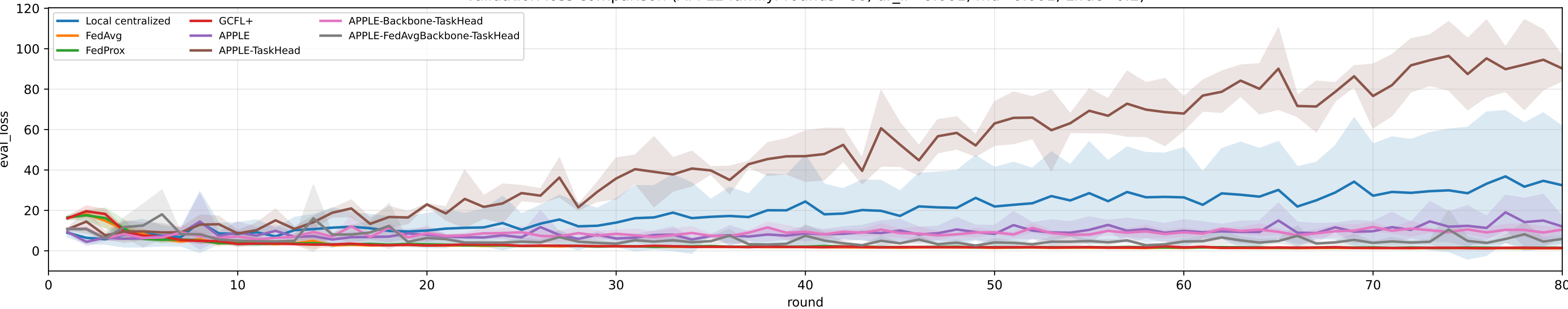
Controlled q-label heterogeneity: High | aggregated / pooled performance over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

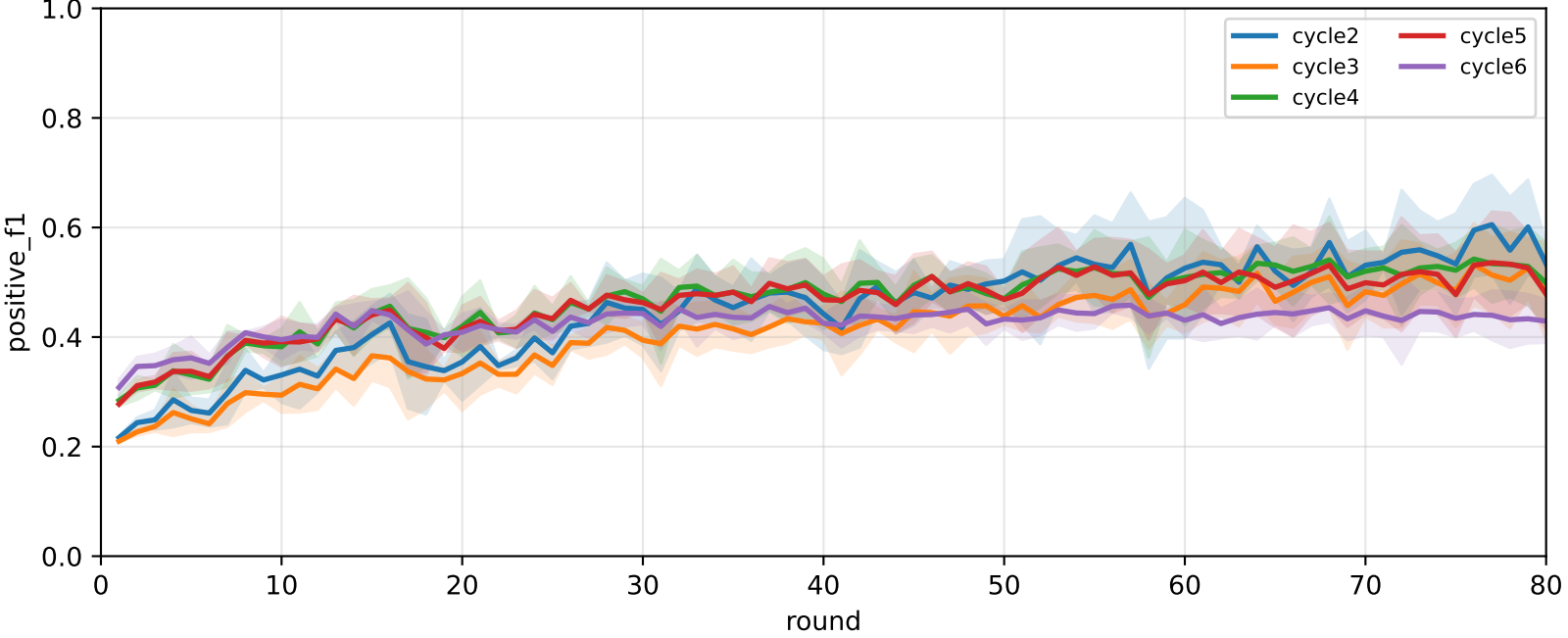
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	29.1% ± 5.1%	29.5% ± 5.3%	22.1% ± 2.8%	26.7% ± 3.4%	26.9% ± 3.1%	26.5% ± 3.3%	30.2% ± 3.0%
FedAvg	54.6% ± 5.2%	55.0% ± 5.3%	50.2% ± 5.4%	60.6% ± 7.5%	56.9% ± 5.9%	57.5% ± 6.2%	49.7% ± 4.8%
FedProx	52.8% ± 2.9%	53.2% ± 2.8%	53.5% ± 2.9%	55.1% ± 3.6%	54.7% ± 5.7%	53.5% ± 6.0%	49.1% ± 1.1%
GCFL+	54.1% ± 1.1%	54.7% ± 1.3%	53.8% ± 4.5%	58.5% ± 4.0%	54.0% ± 4.5%	57.3% ± 5.5%	49.9% ± 0.9%
APPLE	28.0% ± 3.6%	27.9% ± 3.6%	23.0% ± 3.2%	28.3% ± 3.9%	28.2% ± 3.3%	27.4% ± 2.9%	31.0% ± 2.6%
APPLE-TaskHead	31.9% ± 1.7%	31.8% ± 1.7%	26.3% ± 2.0%	31.9% ± 2.1%	32.2% ± 1.8%	32.1% ± 1.4%	34.7% ± 1.0%
APPLE-Backbone-TaskHead	39.7% ± 5.7%	39.7% ± 5.8%	34.1% ± 6.6%	41.0% ± 6.8%	40.3% ± 5.5%	40.7% ± 6.5%	41.0% ± 3.6%
APPLE-FedAvgBackbone-TaskHead	58.8% ± 2.9%	60.2% ± 3.1%	61.5% ± 6.2%	62.7% ± 8.3%	61.0% ± 0.9%	62.9% ± 2.0%	50.0% ± 3.6%

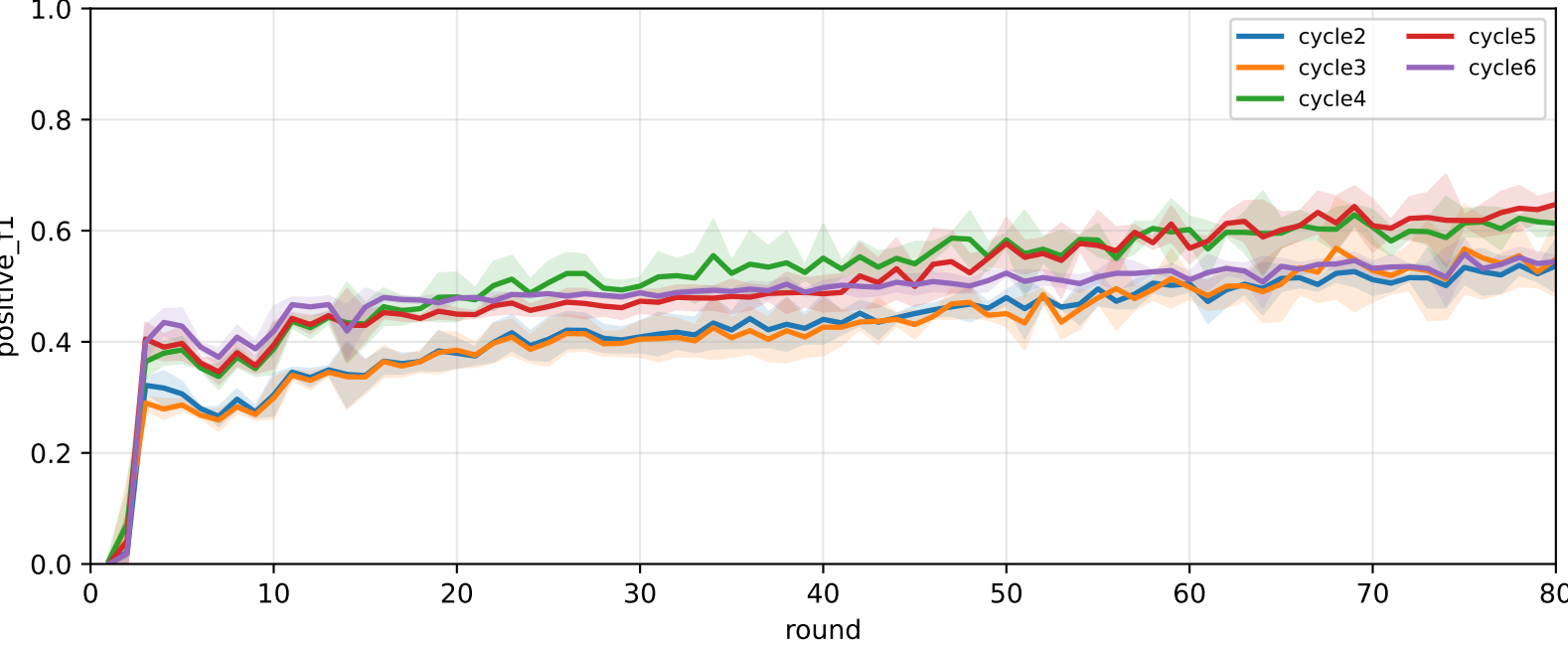
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



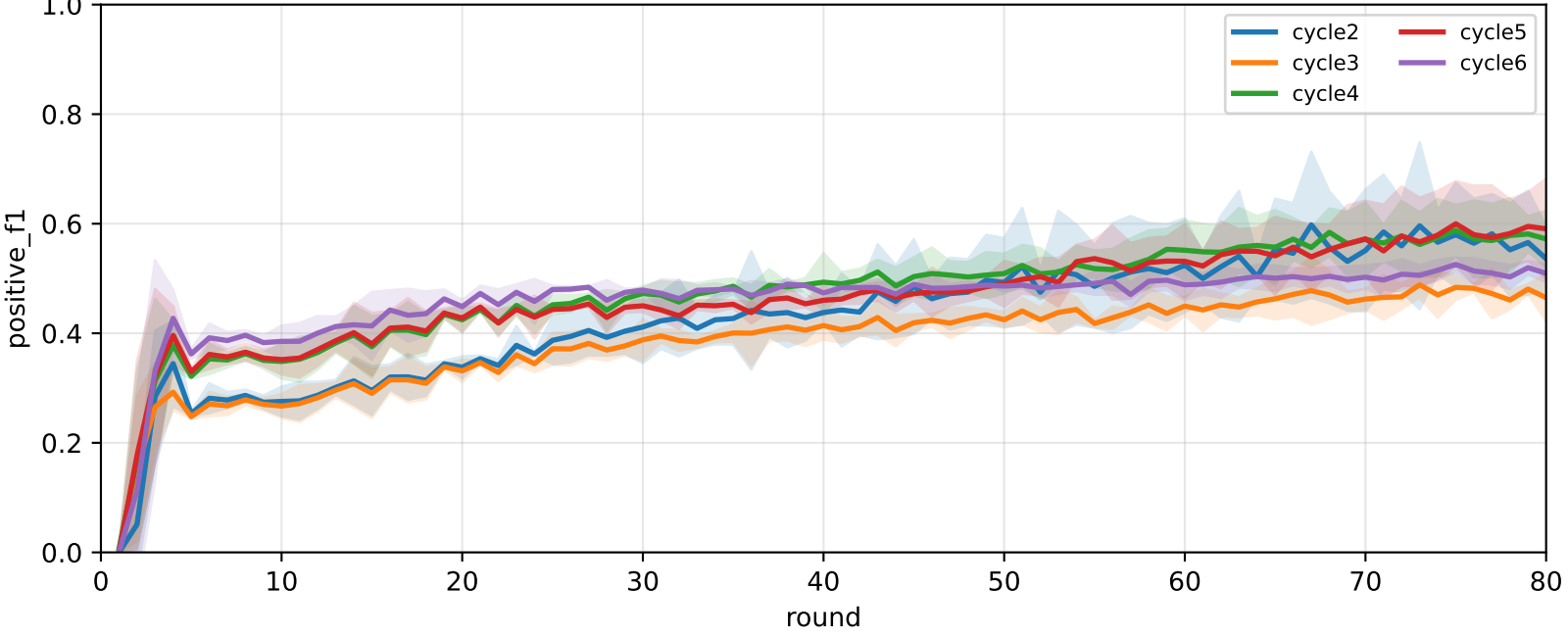
Pooled local centralized positive F1



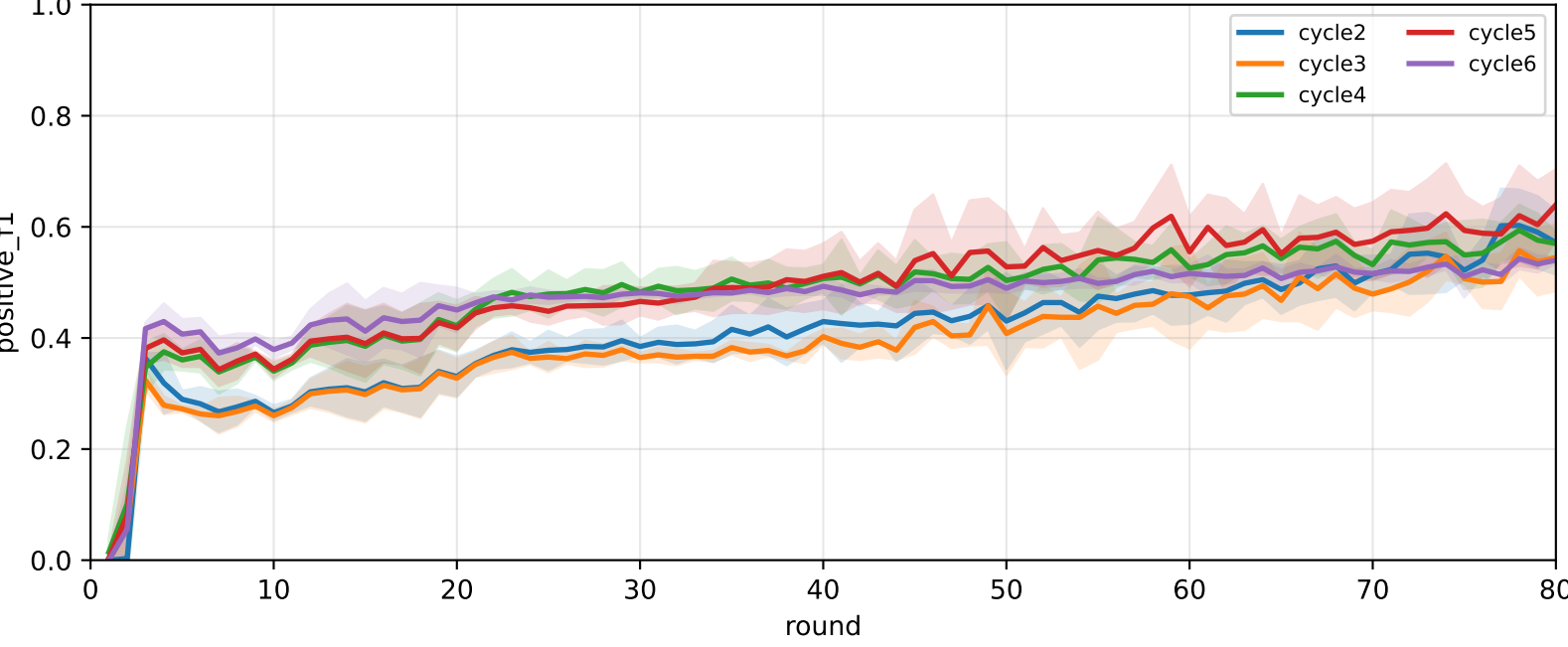
Pooled FedAvg positive F1



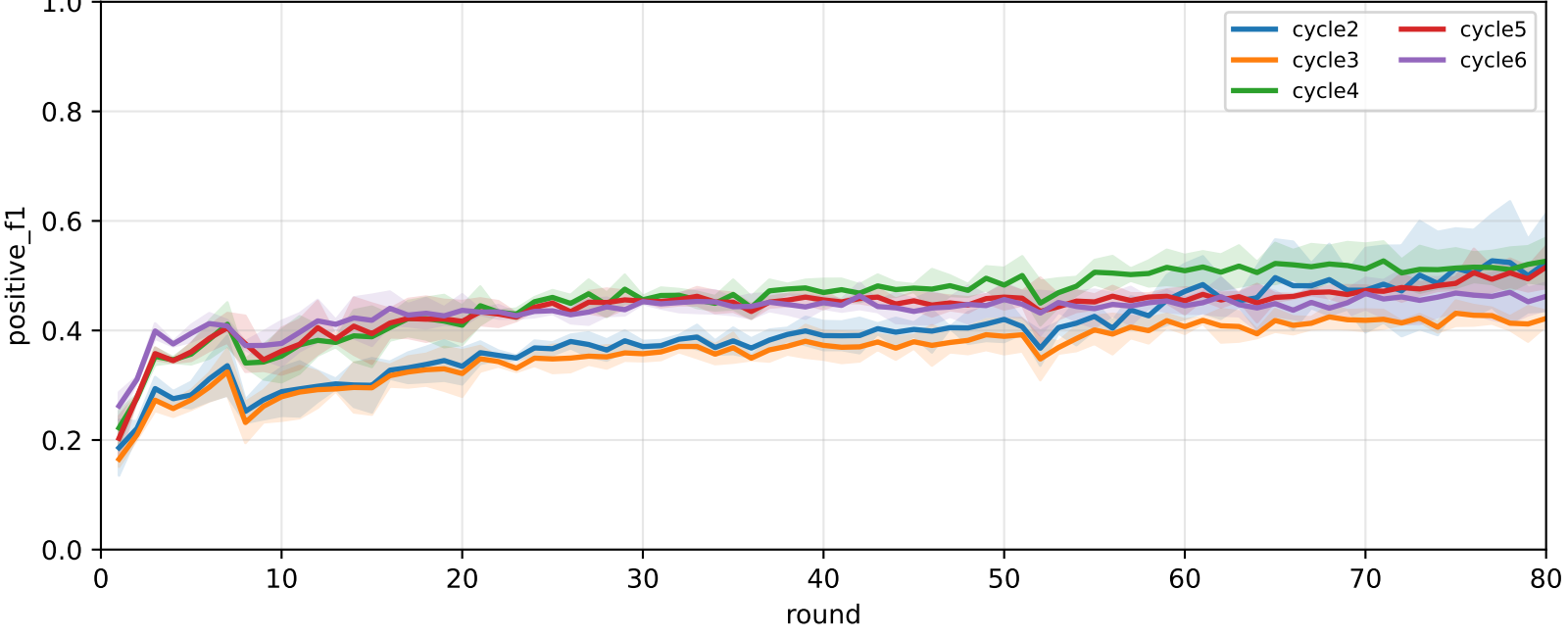
Pooled FedProx positive F1



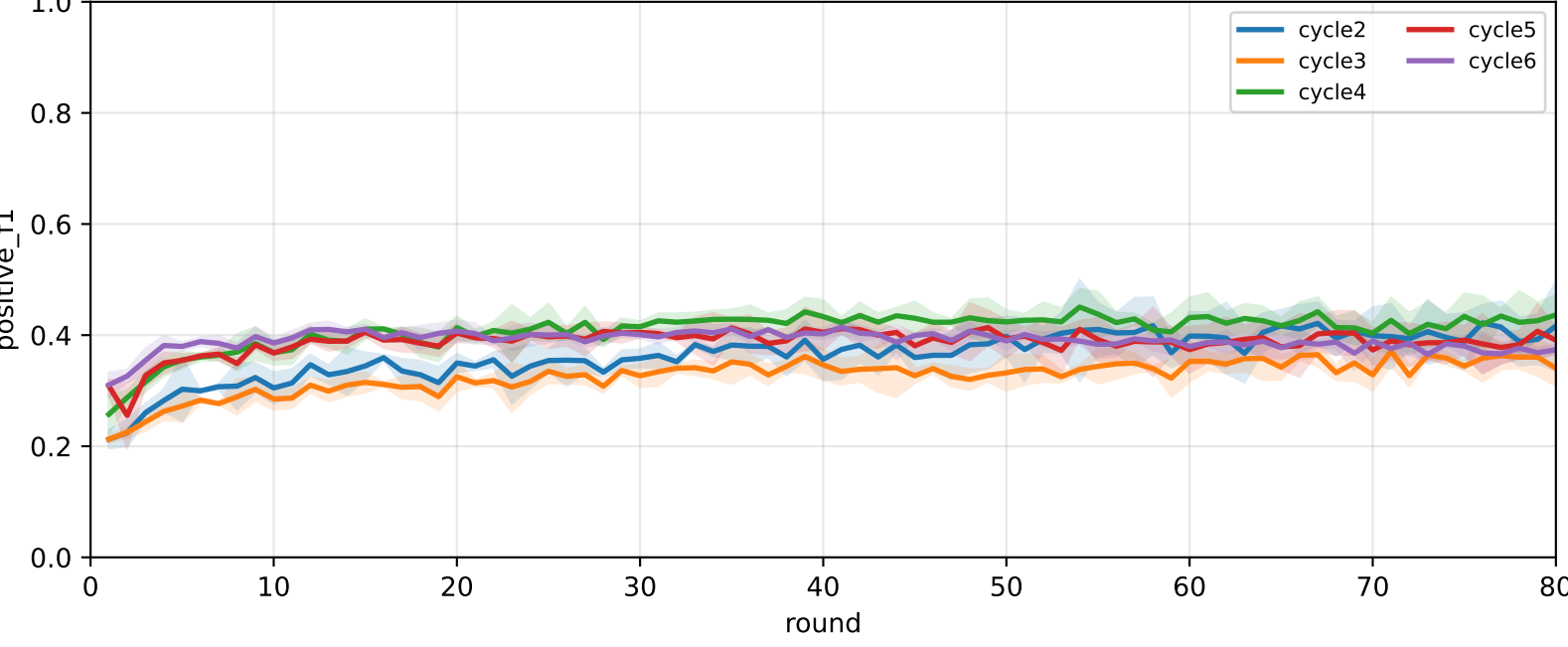
Pooled GCFL+ positive F1



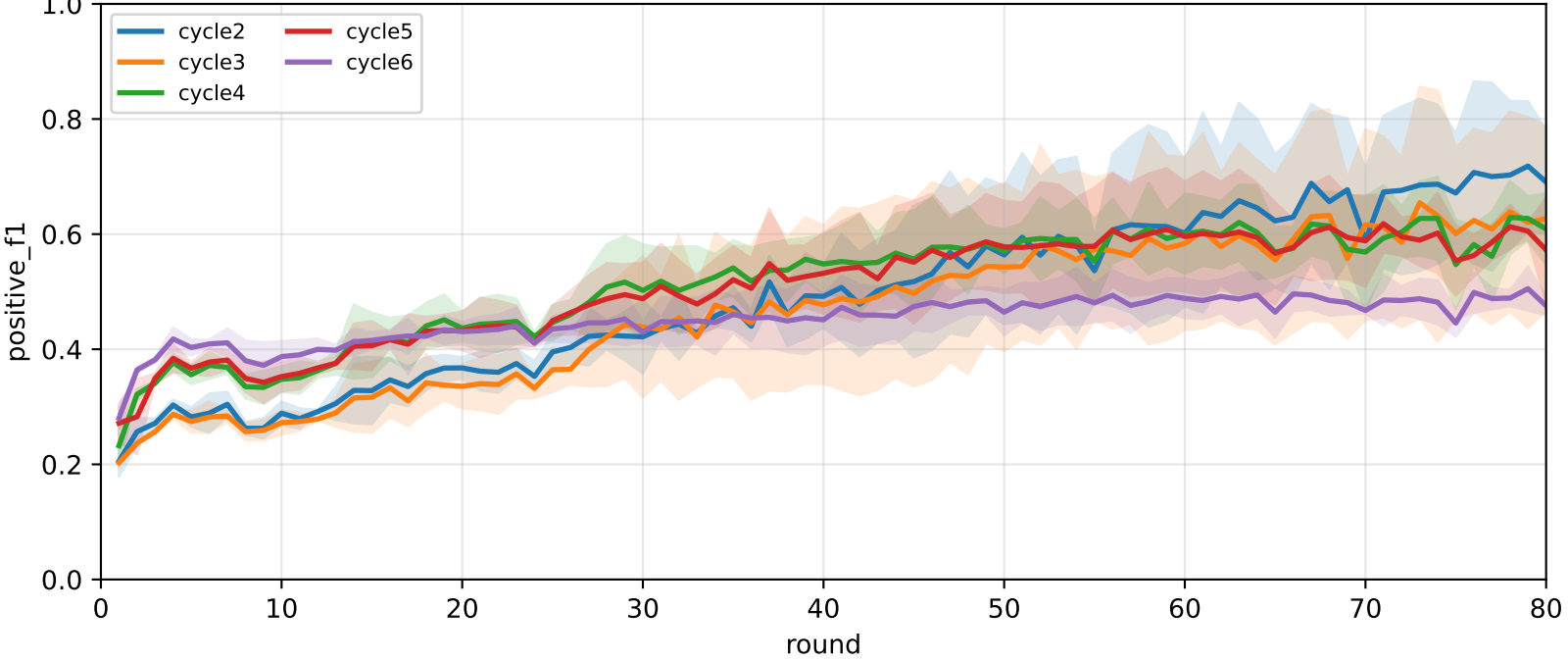
Pooled APPLE positive F1 (after-local selected model, rounds=80)



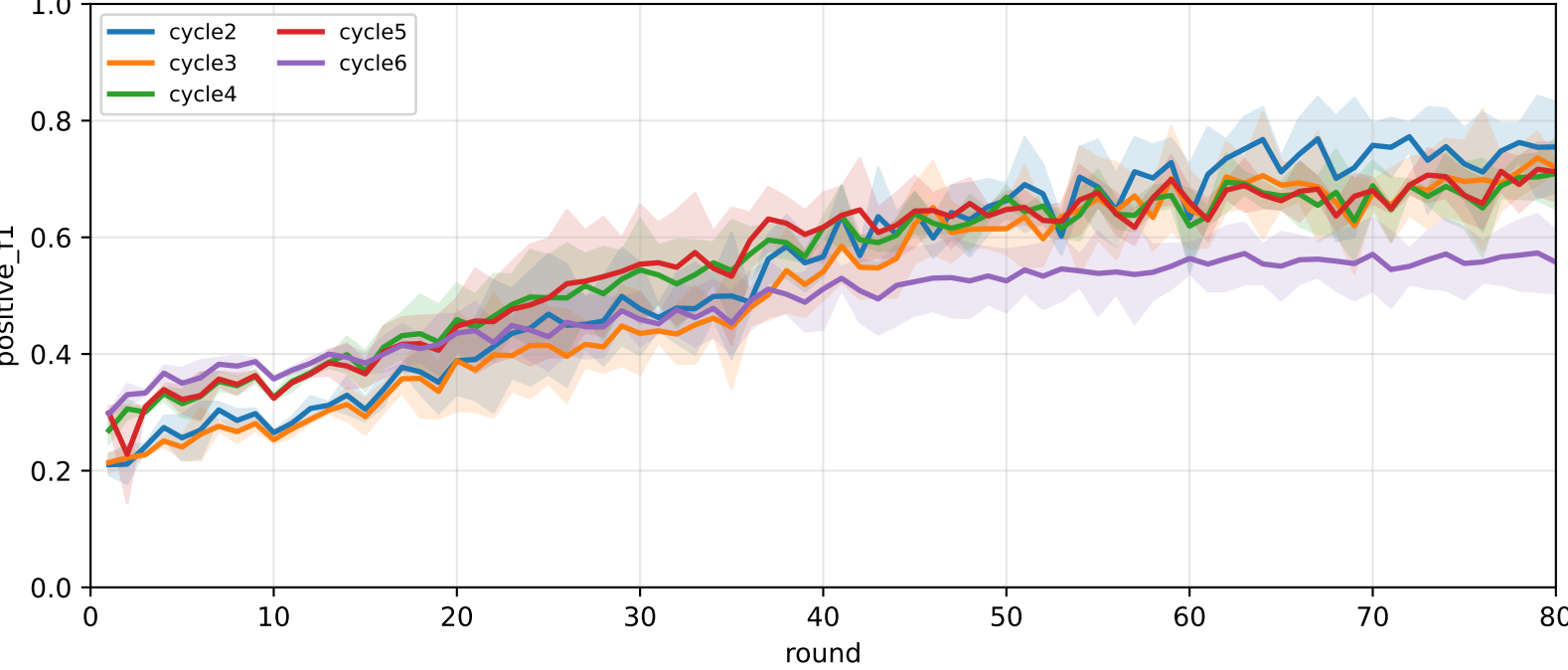
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



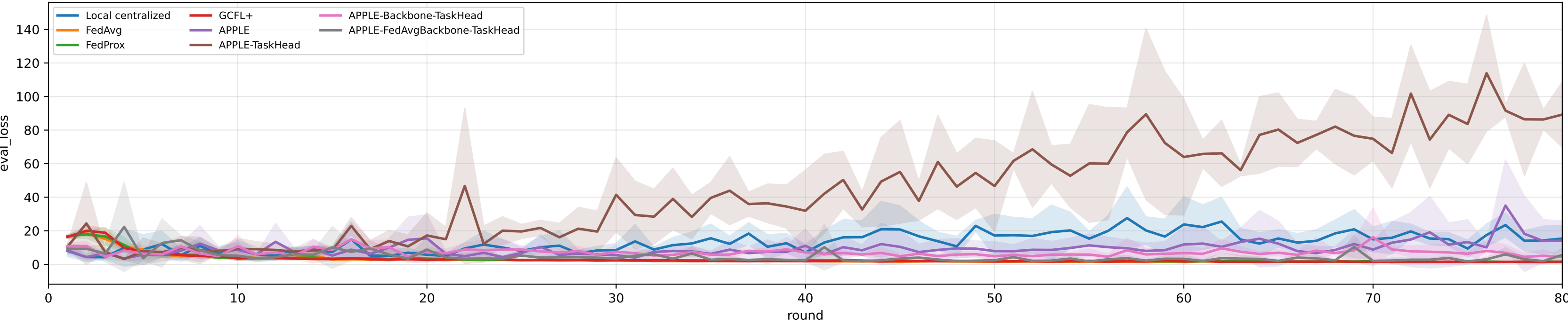
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

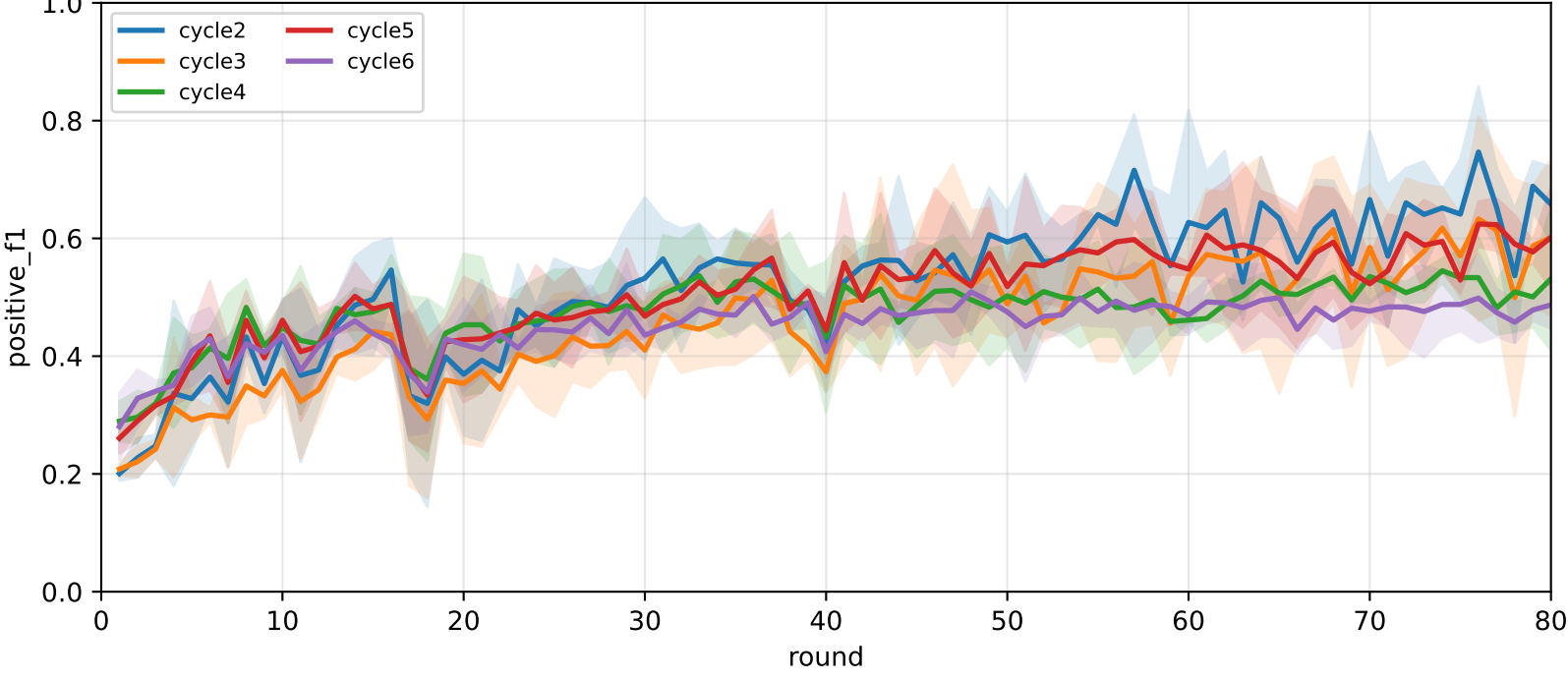
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	32.7% ± 12.1%	33.2% ± 12.8%	31.2% ± 16.3%	35.6% ± 15.5%	35.8% ± 15.4%	31.2% ± 10.3%	32.3% ± 6.7%
FedAvg	54.5% ± 5.3%	55.0% ± 5.4%	50.1% ± 5.9%	60.4% ± 7.5%	57.0% ± 6.0%	57.5% ± 6.1%	49.7% ± 4.8%
FedProx	52.9% ± 2.8%	53.3% ± 2.7%	53.6% ± 3.3%	55.0% ± 3.6%	54.9% ± 5.7%	53.7% ± 6.0%	49.4% ± 1.1%
GCFL+	54.6% ± 1.2%	55.1% ± 1.4%	54.0% ± 4.1%	59.1% ± 4.2%	54.4% ± 4.2%	57.9% ± 5.4%	50.3% ± 0.8%
APPLE	25.7% ± 1.5%	25.6% ± 1.4%	20.9% ± 1.5%	26.1% ± 1.7%	26.2% ± 1.5%	25.8% ± 0.9%	29.2% ± 1.5%
APPLE-TaskHead	33.4% ± 1.0%	33.4% ± 1.0%	27.6% ± 0.6%	34.8% ± 0.8%	34.3% ± 1.3%	33.8% ± 1.3%	36.7% ± 1.2%
APPLE-Backbone-TaskHead	40.9% ± 6.0%	40.9% ± 6.0%	35.0% ± 7.4%	42.5% ± 7.5%	41.3% ± 5.7%	42.8% ± 7.0%	43.0% ± 3.8%
APPLE-FedAvgBackbone-TaskHead	56.8% ± 6.3%	57.6% ± 6.0%	53.8% ± 7.4%	62.7% ± 10.2%	60.7% ± 5.6%	59.9% ± 6.9%	50.8% ± 6.9%

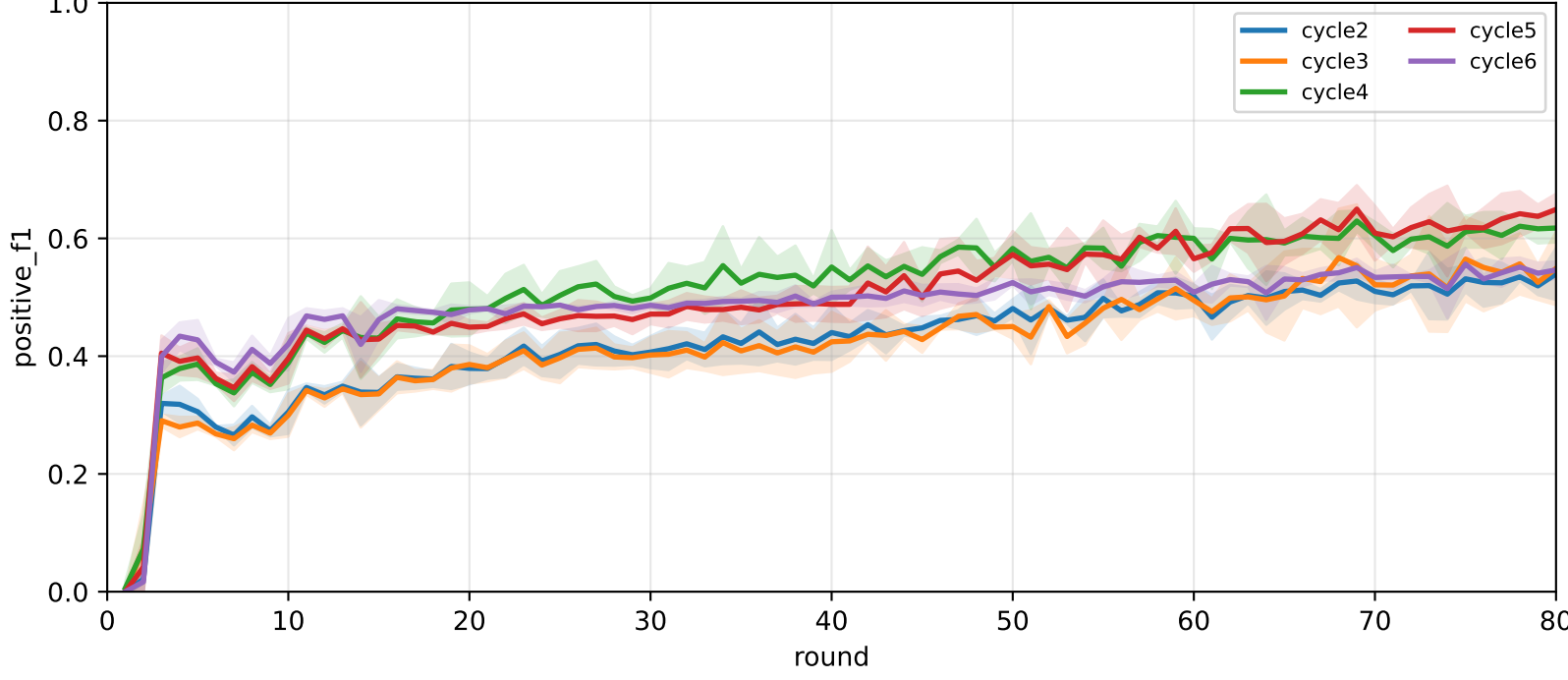
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



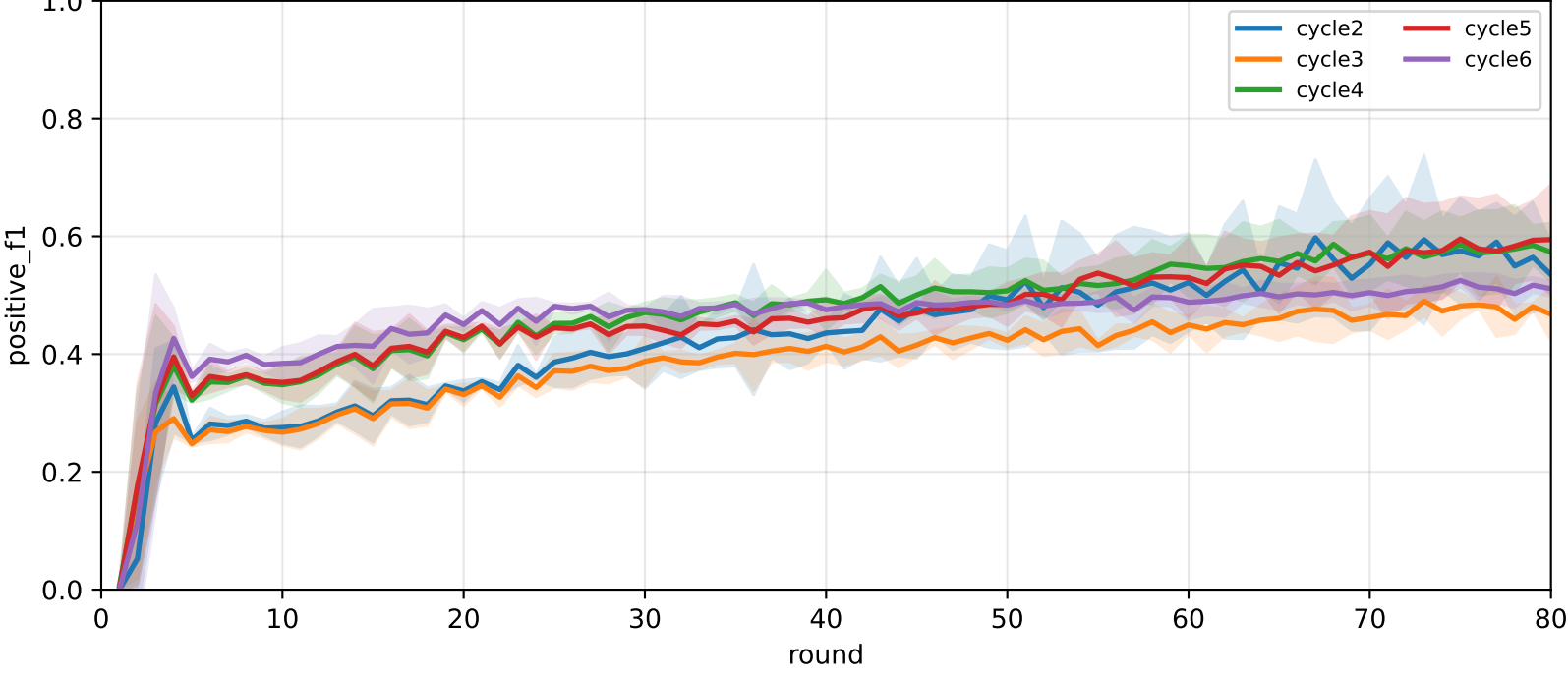
Pooled local centralized positive F1



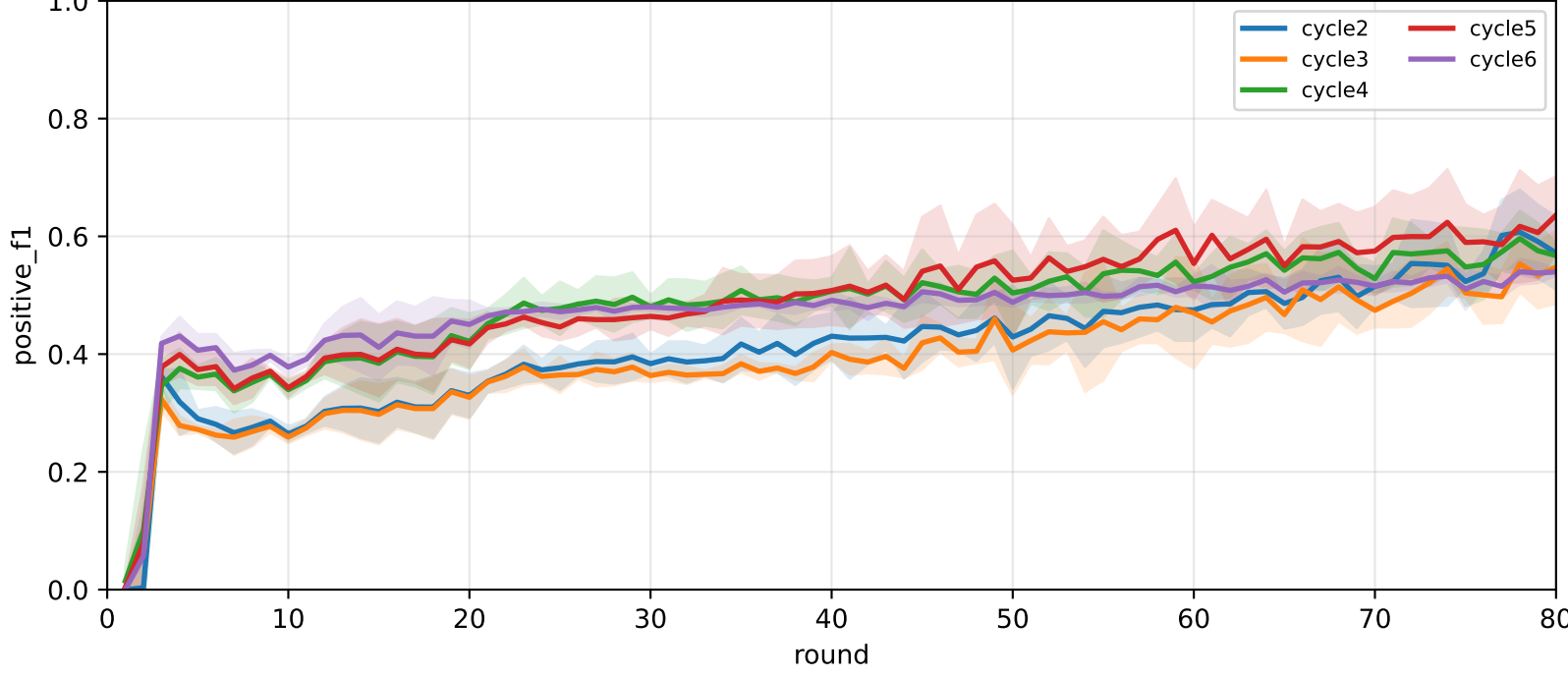
Pooled FedAvg positive F1



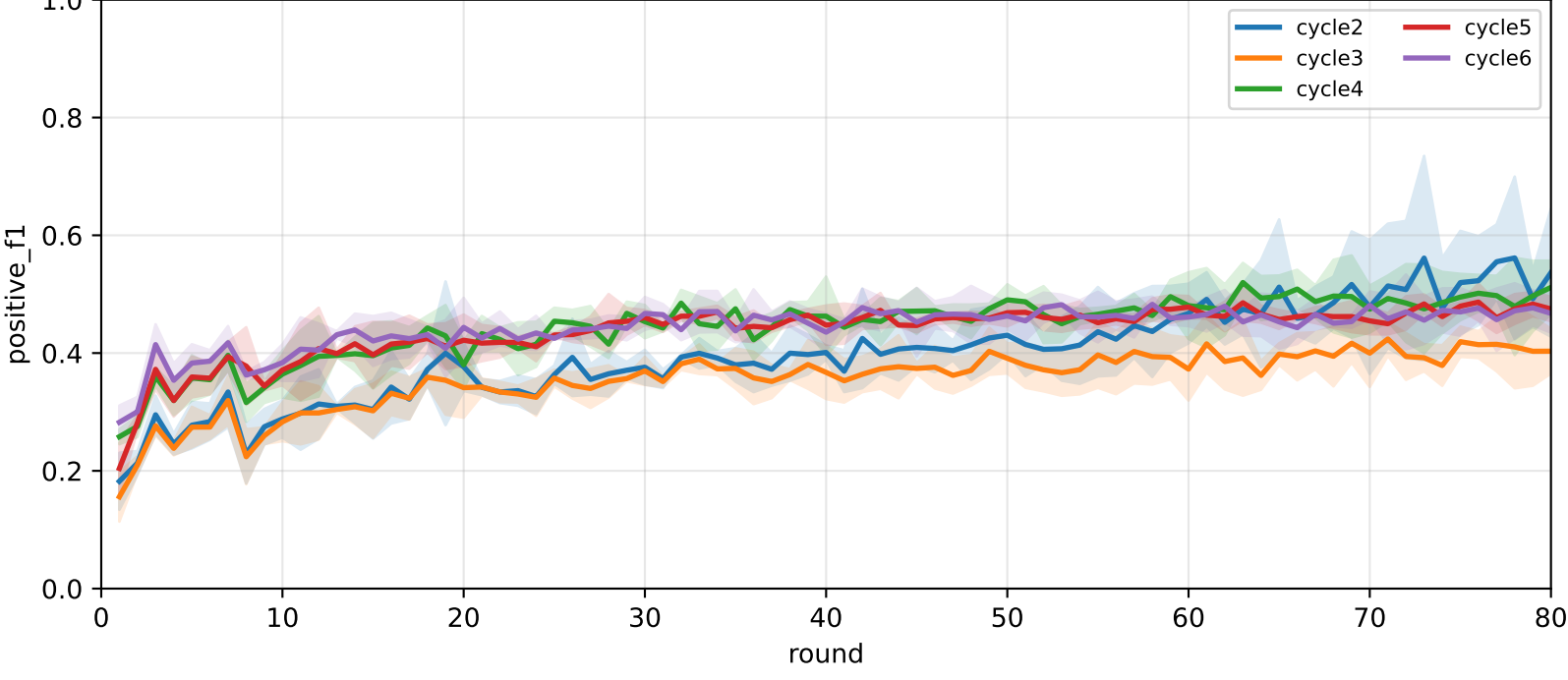
Pooled FedProx positive F1



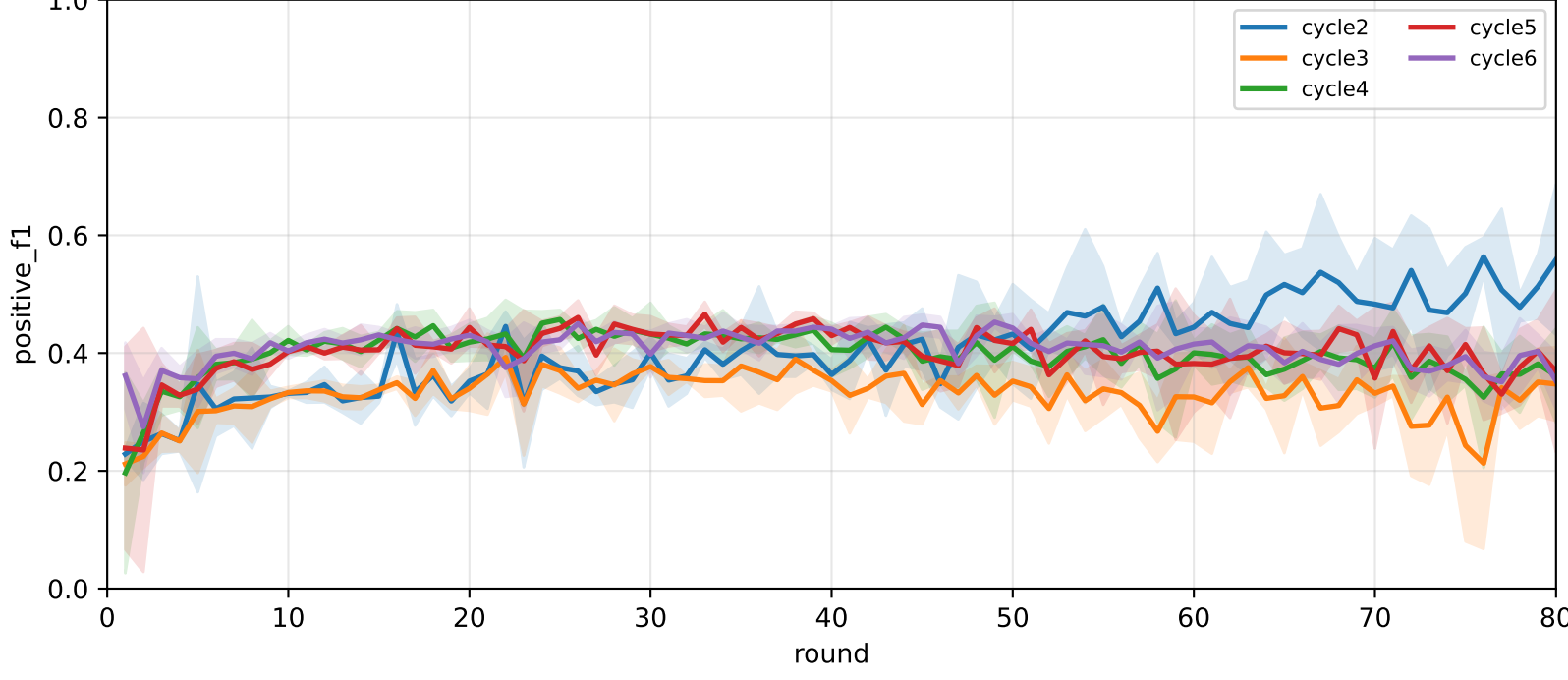
Pooled GCFL+ positive F1



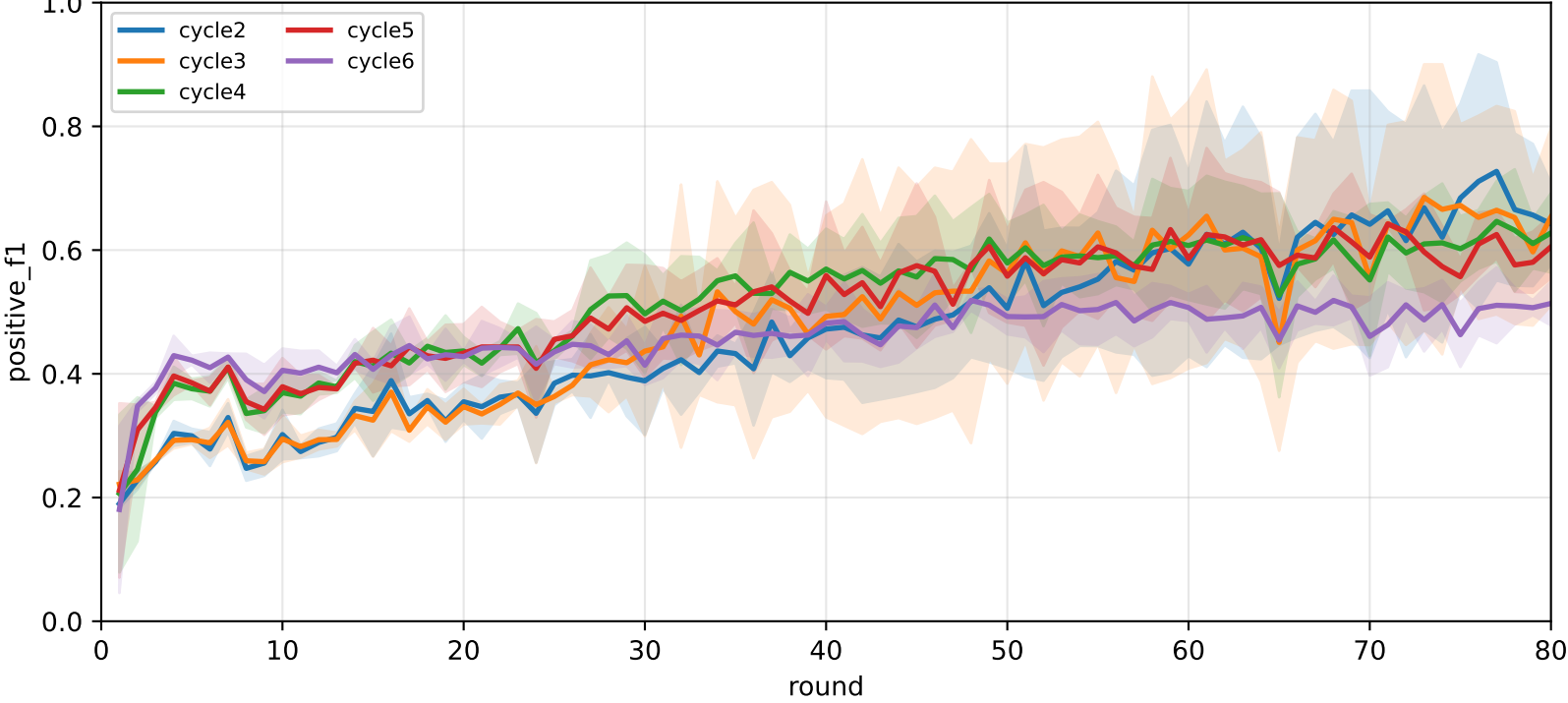
Pooled APPLE positive F1 (after-local selected model, rounds=80)



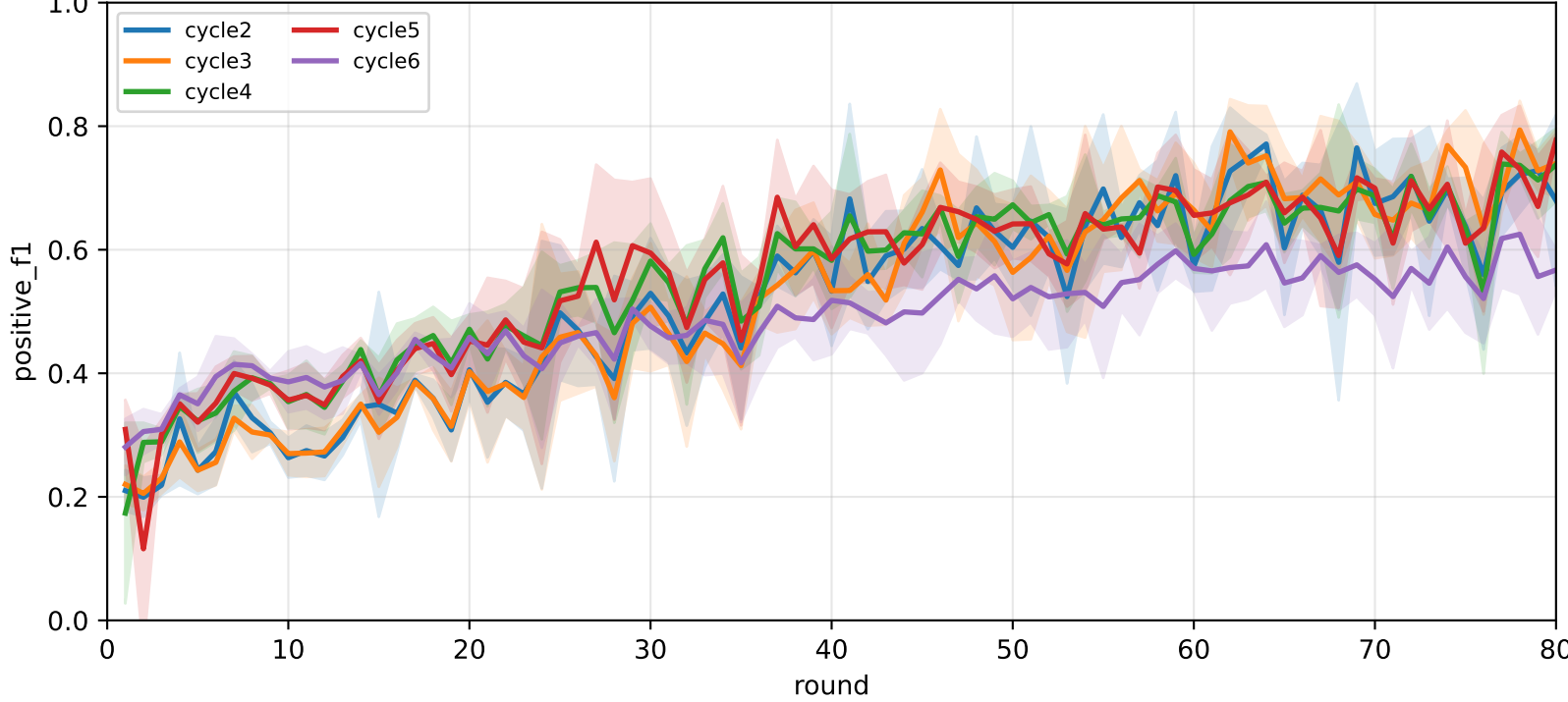
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



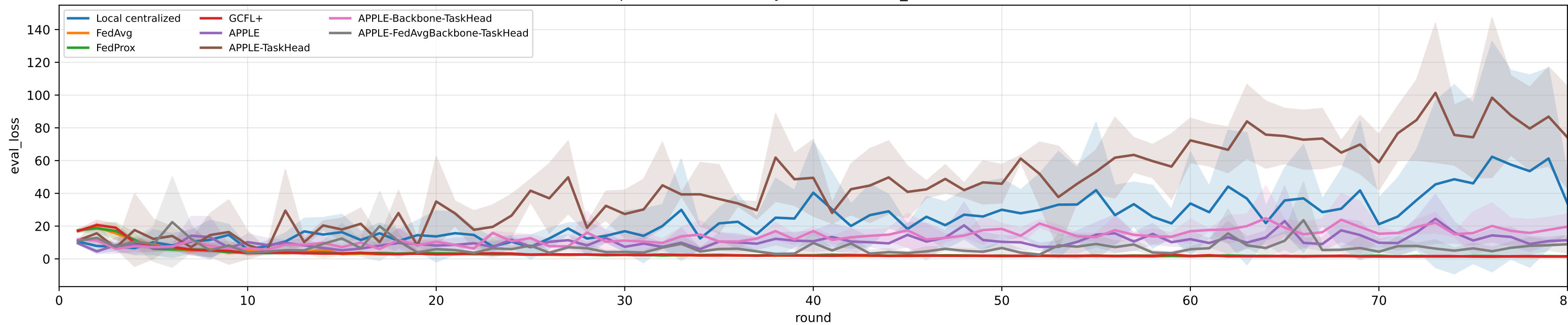
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

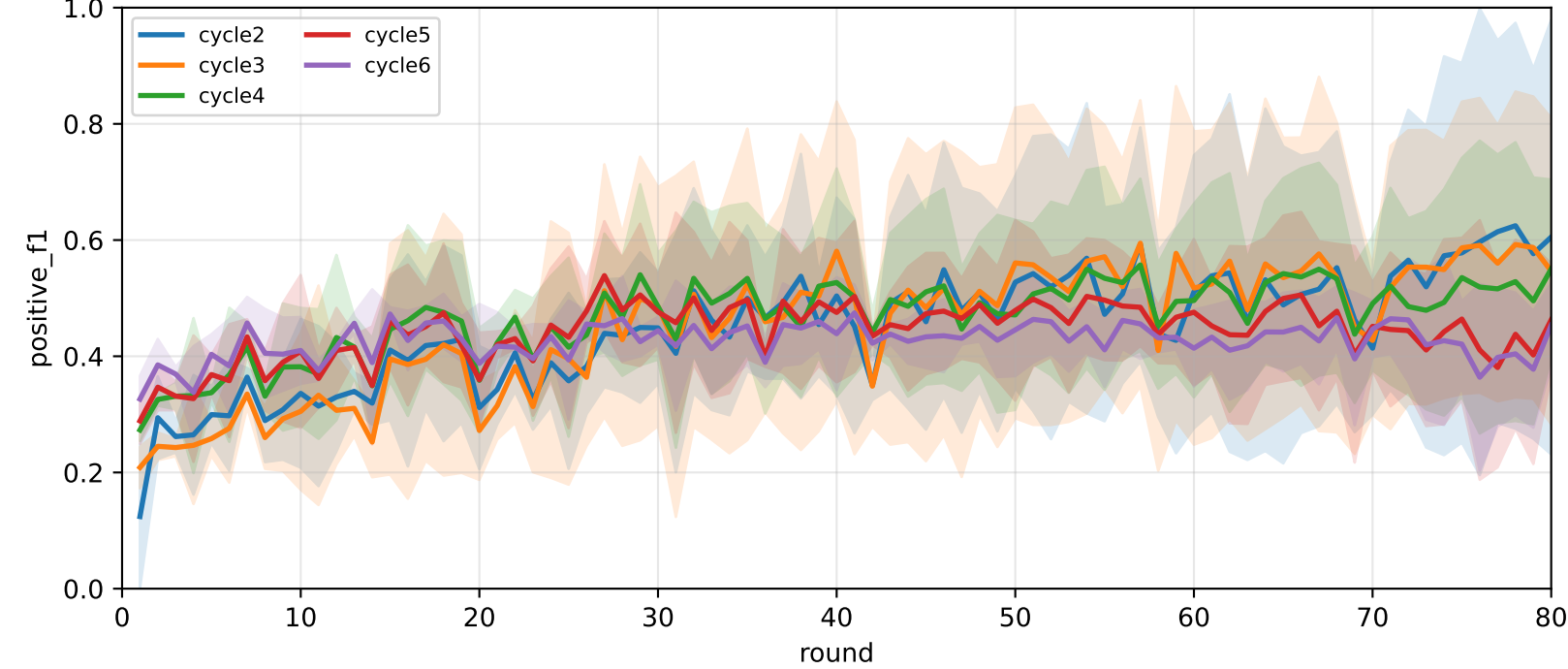
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	24.6% ± 3.5%	24.6% ± 3.5%	20.2% ± 3.0%	23.5% ± 4.2%	24.7% ± 3.6%	25.1% ± 3.4%	29.3% ± 3.3%
FedAvg	54.6% ± 5.7%	55.1% ± 5.8%	50.3% ± 5.8%	60.9% ± 8.5%	56.9% ± 6.3%	57.5% ± 6.3%	49.9% ± 5.1%
FedProx	52.8% ± 2.6%	53.2% ± 2.5%	53.4% ± 3.2%	55.1% ± 3.4%	54.8% ± 5.3%	53.4% ± 5.6%	49.2% ± 0.9%
GCFL+	54.3% ± 1.3%	54.9% ± 1.4%	54.3% ± 4.4%	58.9% ± 4.0%	54.1% ± 5.0%	57.5% ± 5.9%	49.6% ± 0.7%
APPLE	30.0% ± 6.7%	29.9% ± 6.7%	25.8% ± 7.5%	31.1% ± 7.9%	30.4% ± 7.4%	29.9% ± 6.1%	32.6% ± 4.7%
APPLE-TaskHead	34.4% ± 2.6%	34.4% ± 2.6%	29.2% ± 4.8%	34.1% ± 2.8%	36.0% ± 3.5%	35.3% ± 1.3%	37.6% ± 2.0%
APPLE-Backbone-TaskHead	40.8% ± 7.1%	41.0% ± 7.4%	36.0% ± 8.3%	40.2% ± 6.5%	41.7% ± 7.3%	46.7% ± 13.2%	40.6% ± 2.7%
APPLE-FedAvgBackbone-TaskHead	59.9% ± 4.4%	61.5% ± 4.8%	68.8% ± 7.2%	58.2% ± 8.1%	64.8% ± 5.7%	66.1% ± 6.9%	49.8% ± 0.5%

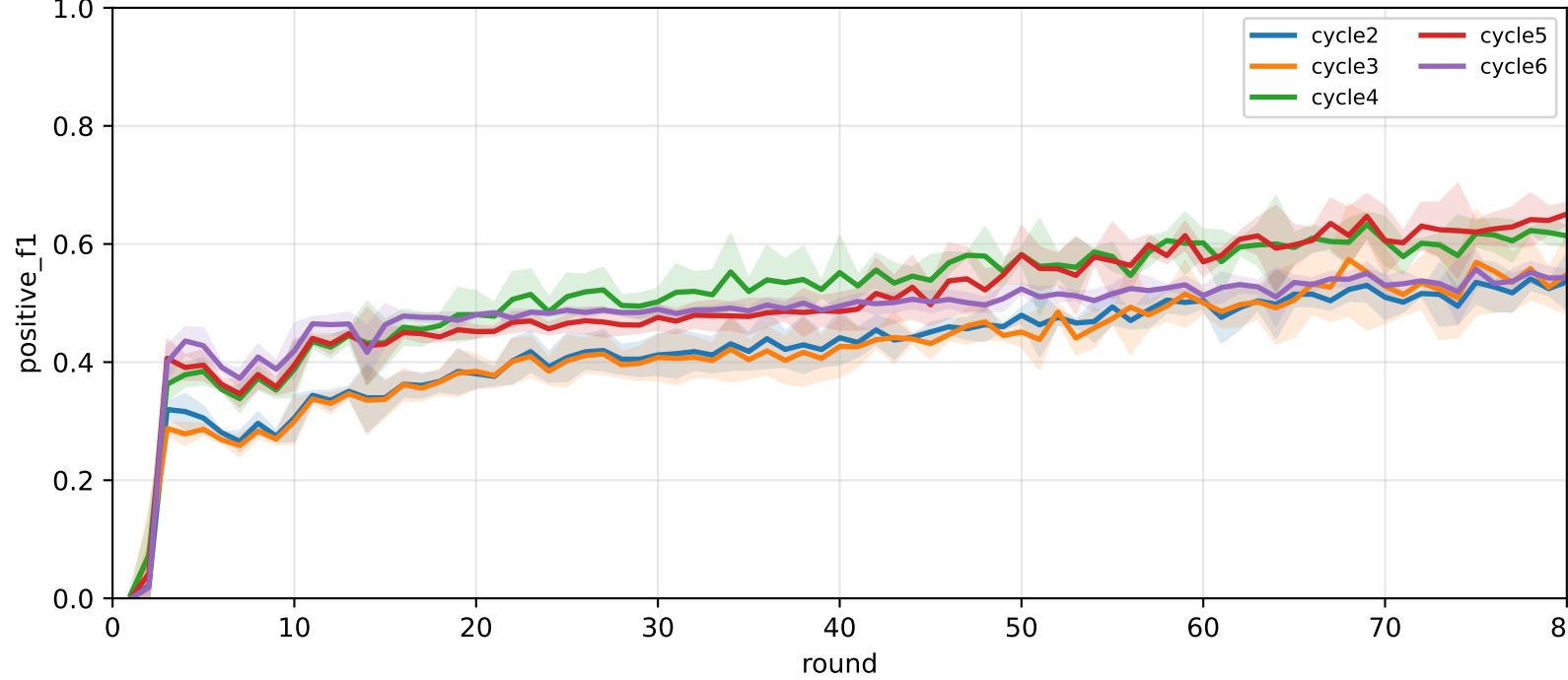
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



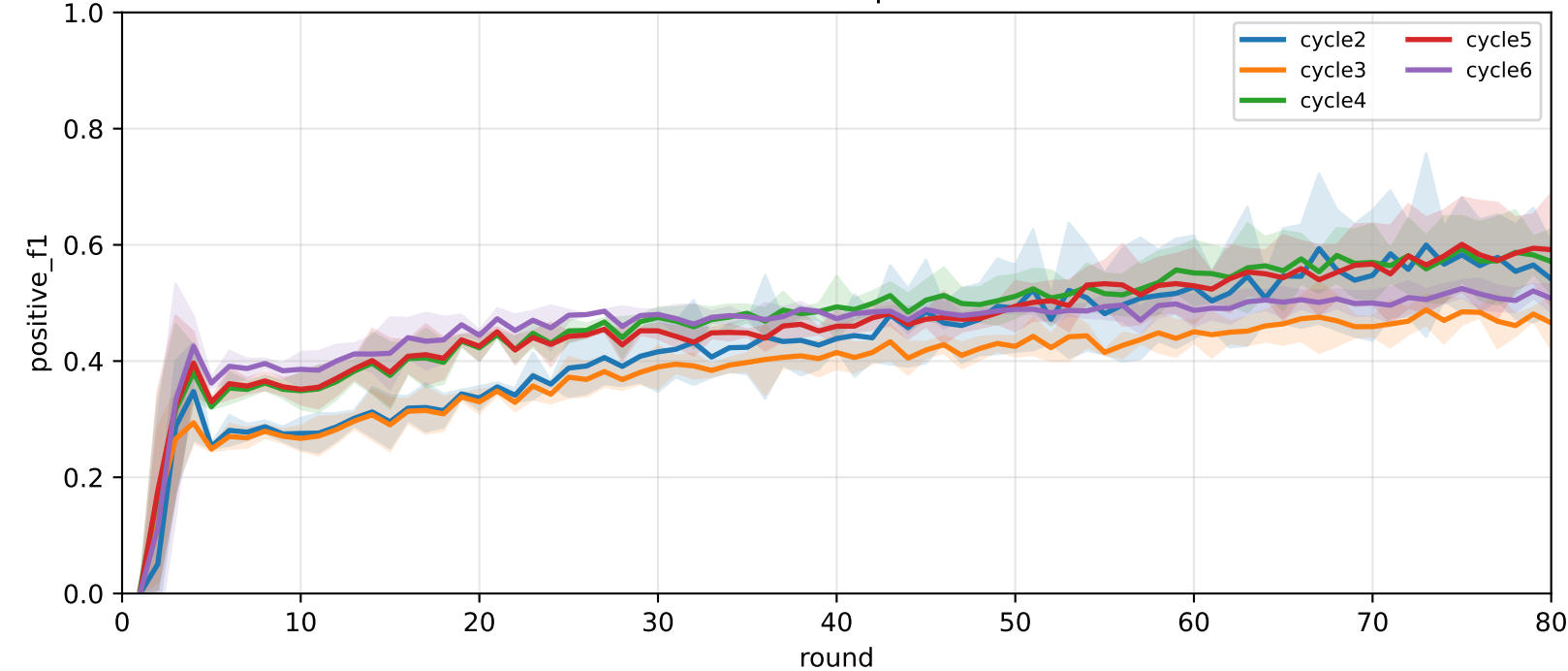
Pooled local centralized positive F1



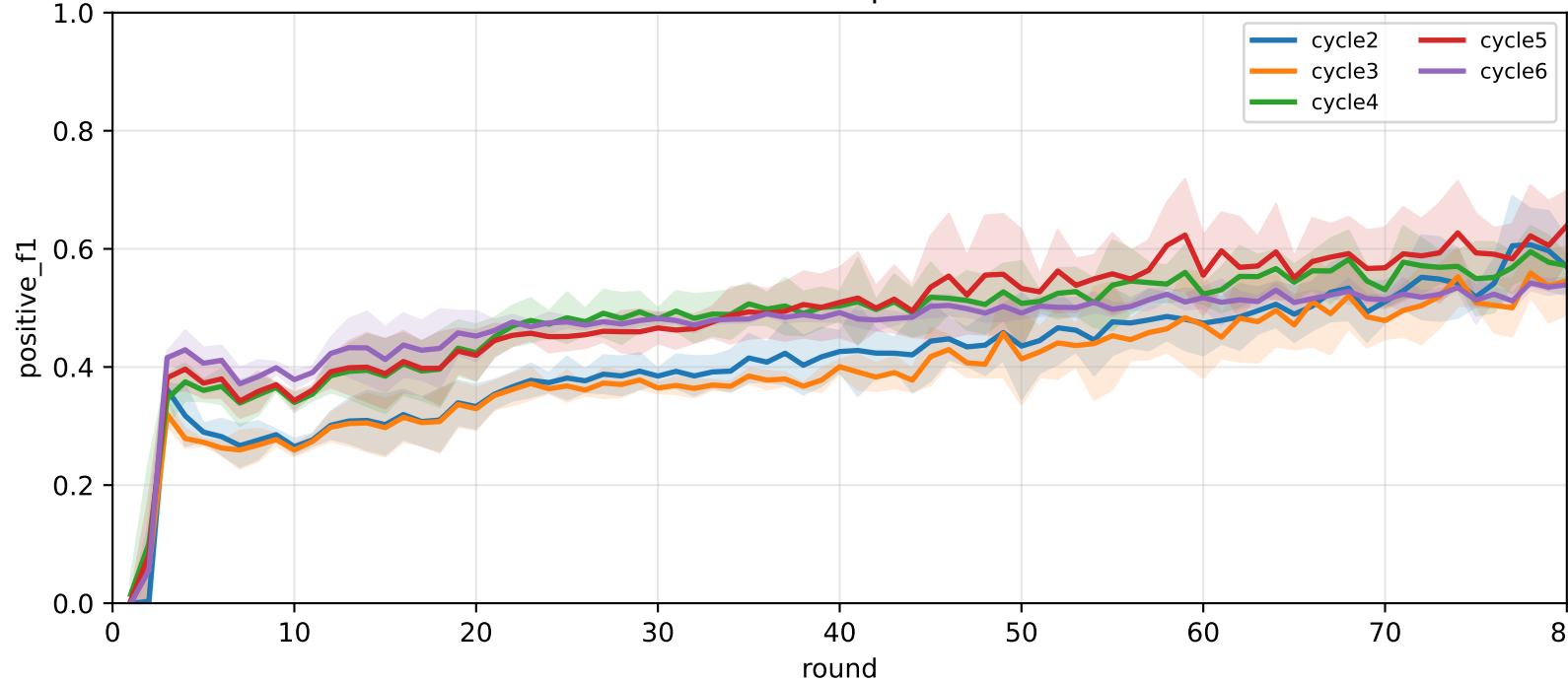
Pooled FedAvg positive F1



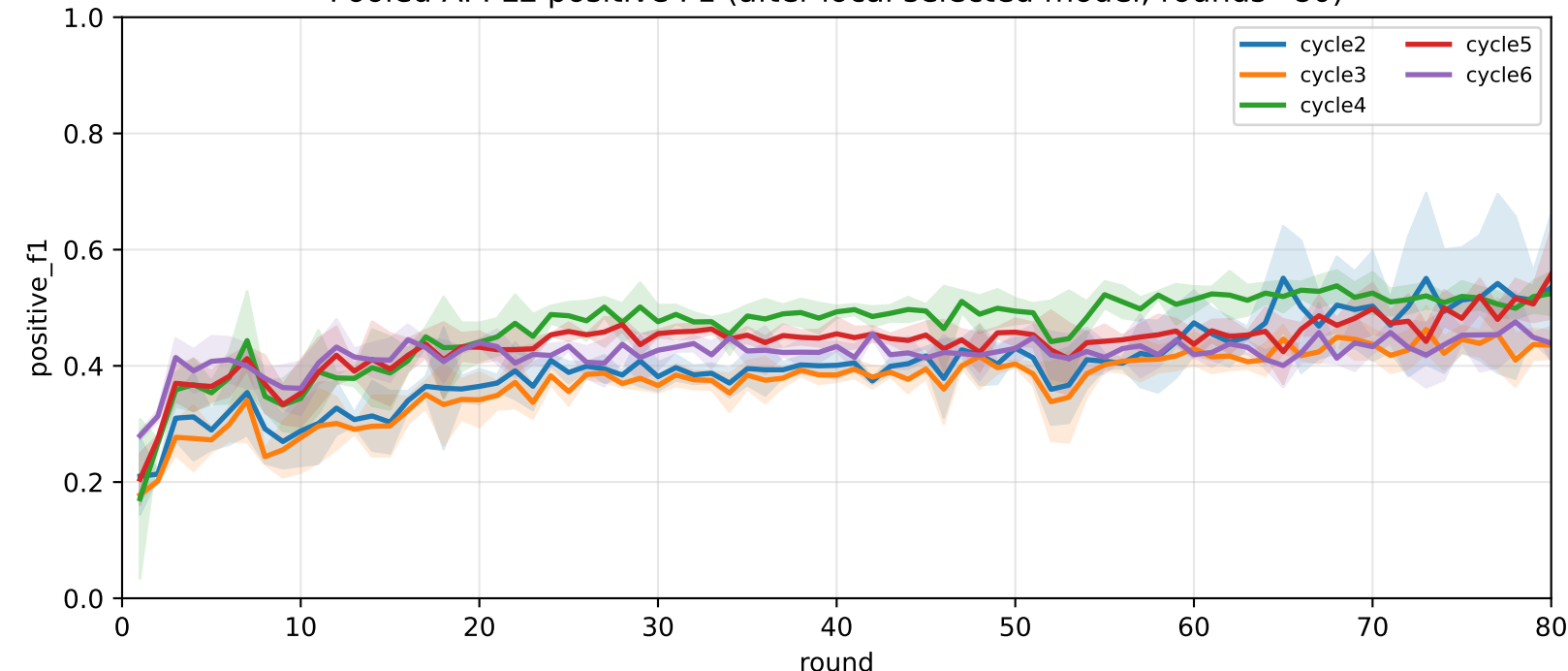
Pooled FedProx positive F1



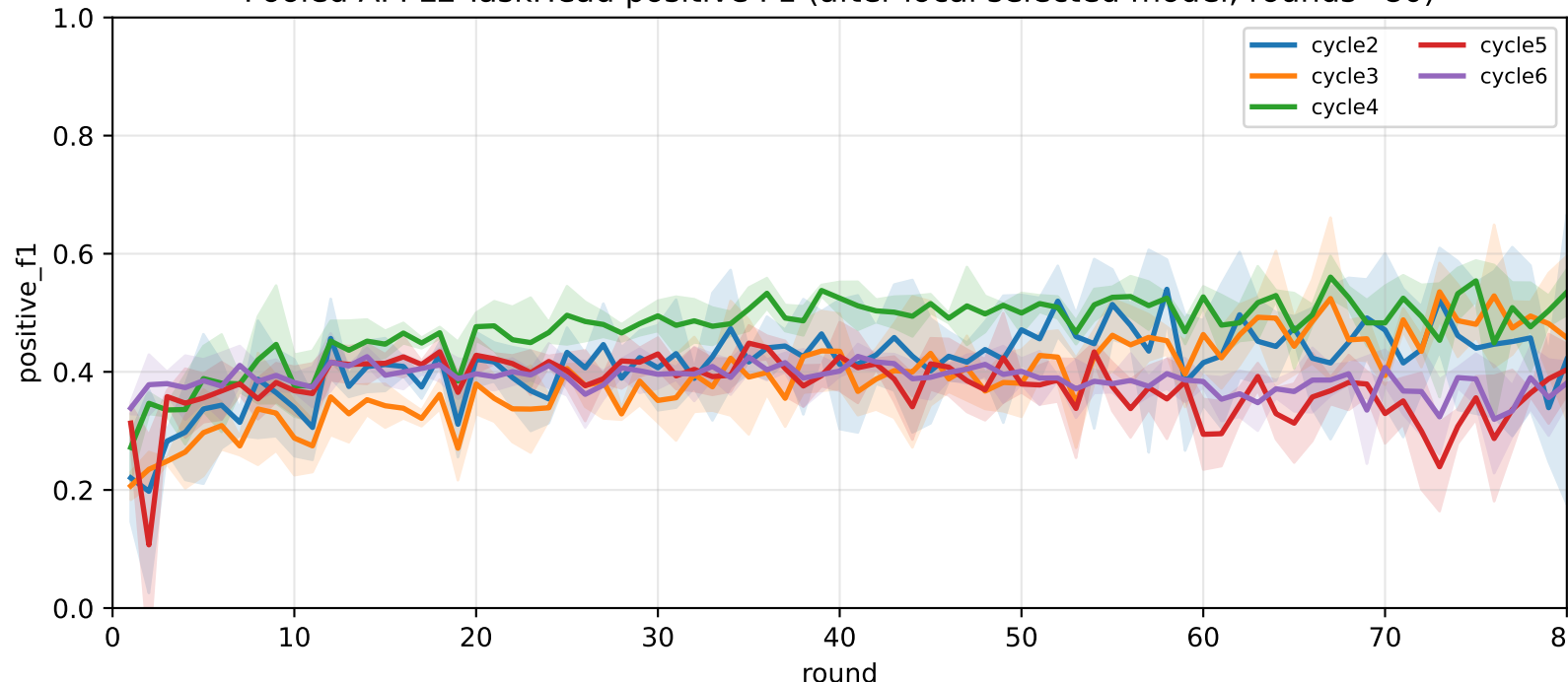
Pooled GCFL+ positive F1



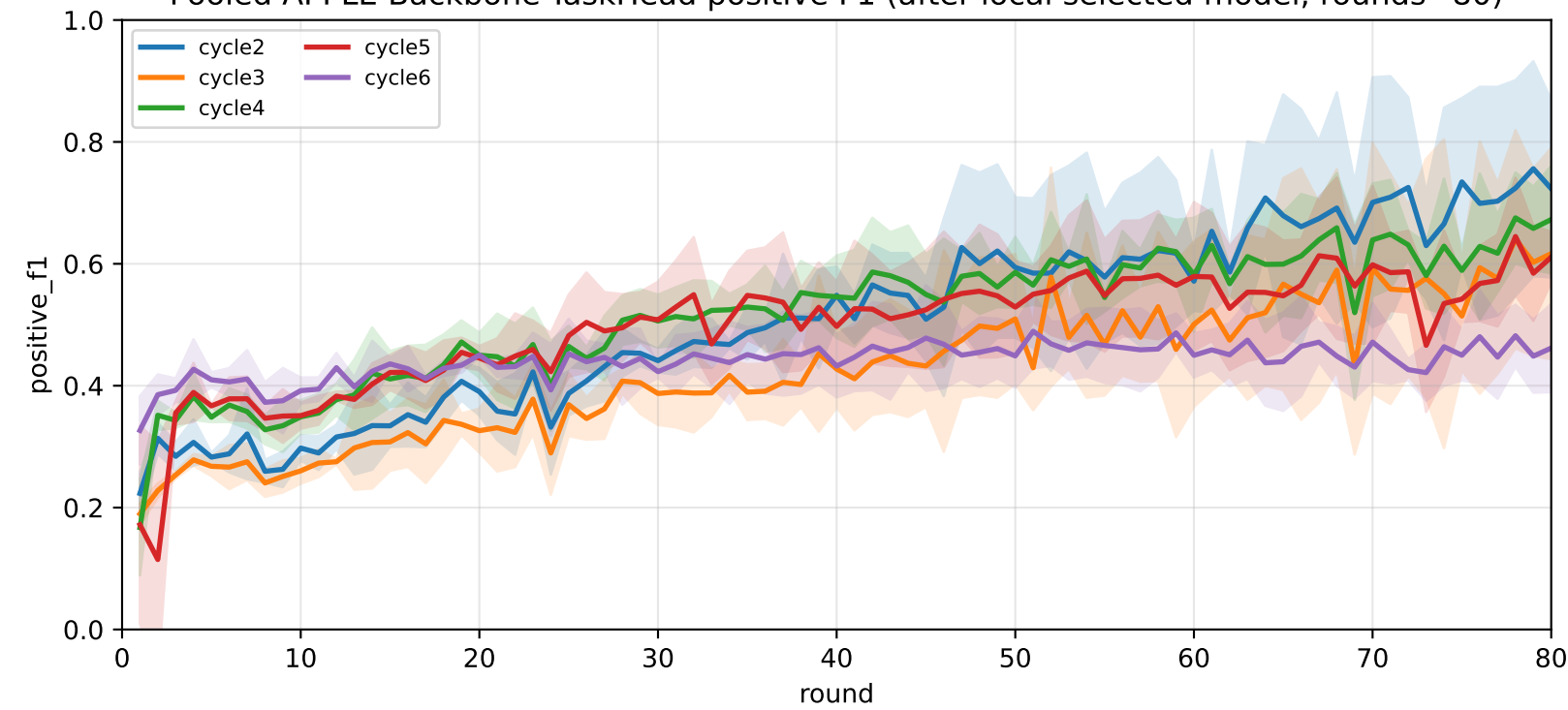
Pooled APPLE positive F1 (after-local selected model, rounds=80)



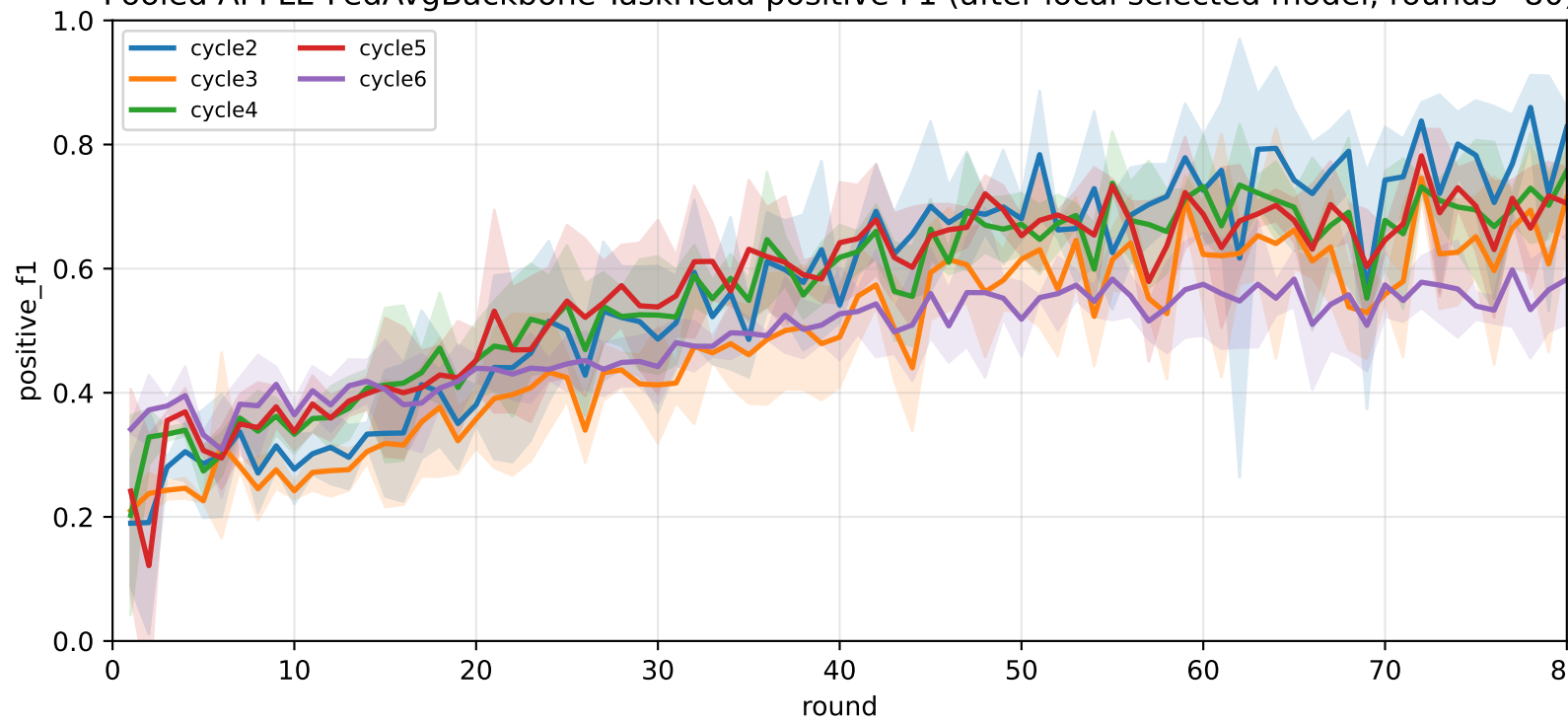
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



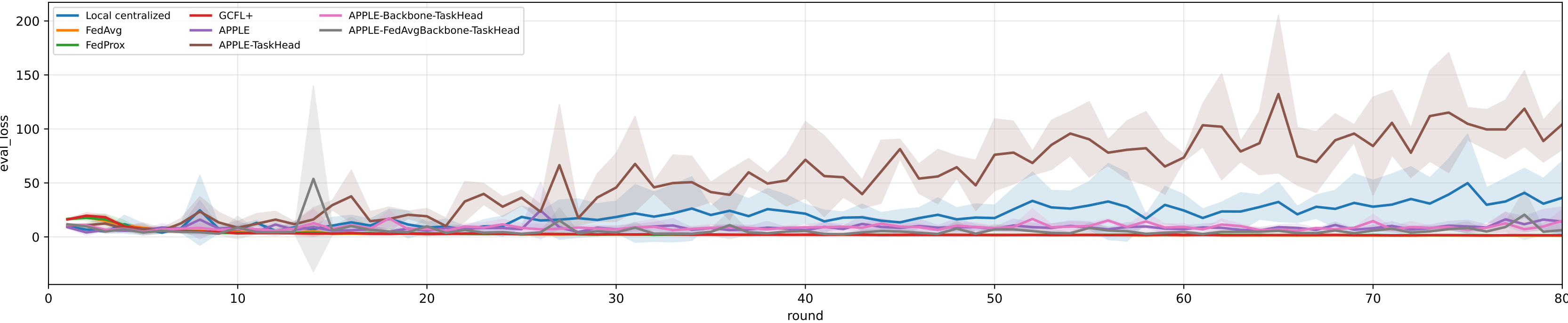
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

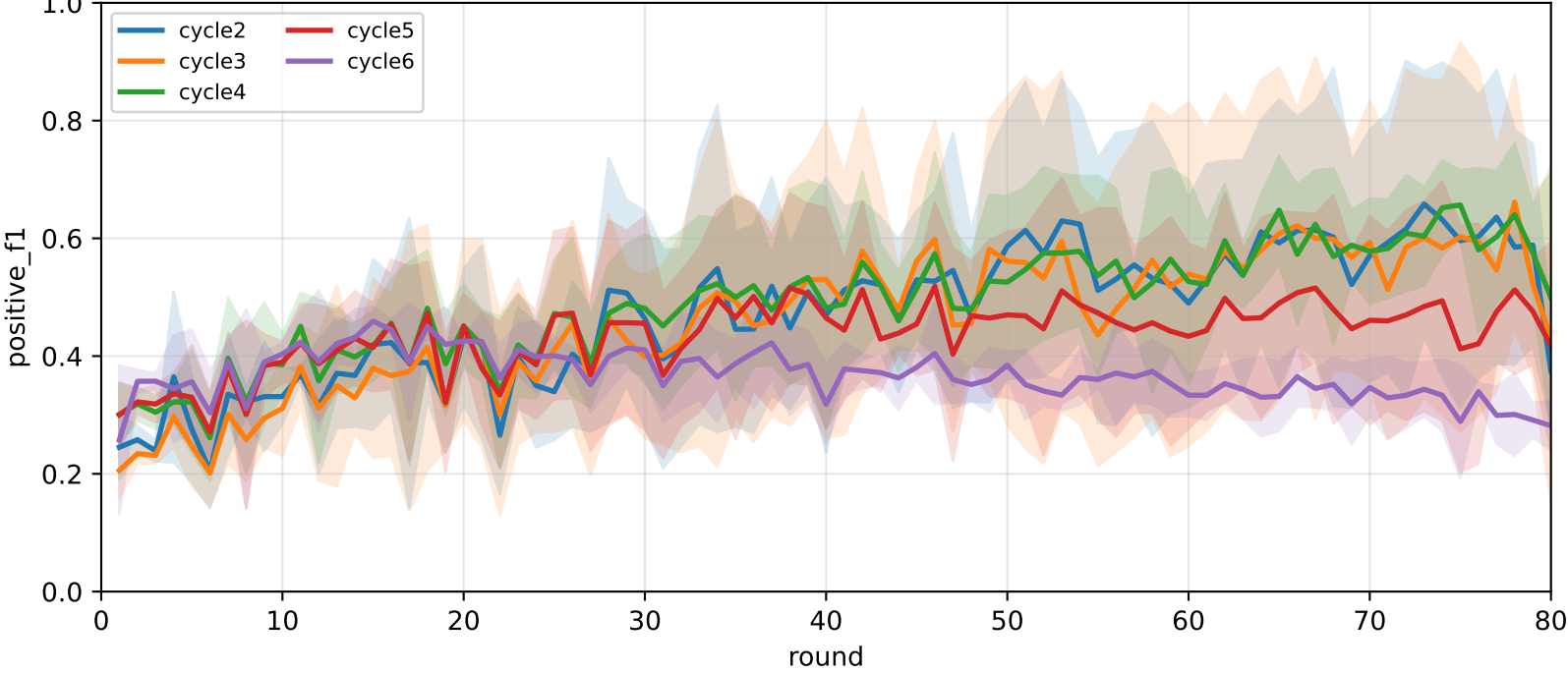
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	20.9% ± 2.3%	21.0% ± 2.5%	16.7% ± 1.9%	21.0% ± 1.6%	20.9% ± 2.8%	20.3% ± 1.5%	26.1% ± 4.9%
FedAvg	54.7% ± 5.4%	55.2% ± 5.5%	50.3% ± 5.6%	60.7% ± 7.2%	57.1% ± 6.4%	58.0% ± 7.1%	49.8% ± 4.9%
FedProx	53.0% ± 2.8%	53.4% ± 2.8%	53.9% ± 3.0%	55.3% ± 3.6%	55.0% ± 5.8%	53.7% ± 6.0%	49.4% ± 0.9%
GCFL+	53.9% ± 1.2%	54.5% ± 1.3%	53.5% ± 4.6%	58.4% ± 4.6%	53.6% ± 4.7%	57.2% ± 5.5%	49.8% ± 1.0%
APPLE	29.0% ± 5.5%	29.0% ± 5.5%	25.0% ± 5.6%	29.6% ± 6.6%	29.4% ± 5.9%	28.5% ± 5.1%	32.5% ± 4.3%
APPLE-TaskHead	29.0% ± 0.9%	29.0% ± 0.9%	24.3% ± 0.6%	29.2% ± 0.7%	28.6% ± 1.0%	30.1% ± 1.8%	32.6% ± 0.6%
APPLE-Backbone-TaskHead	36.9% ± 4.7%	36.8% ± 4.7%	31.4% ± 4.5%	38.2% ± 5.2%	36.9% ± 5.8%	36.9% ± 4.7%	40.6% ± 4.3%
APPLE-FedAvgBackbone-TaskHead	60.6% ± 4.8%	62.1% ± 5.8%	65.2% ± 10.2%	66.4% ± 12.9%	56.8% ± 4.0%	70.1% ± 3.5%	52.0% ± 2.8%

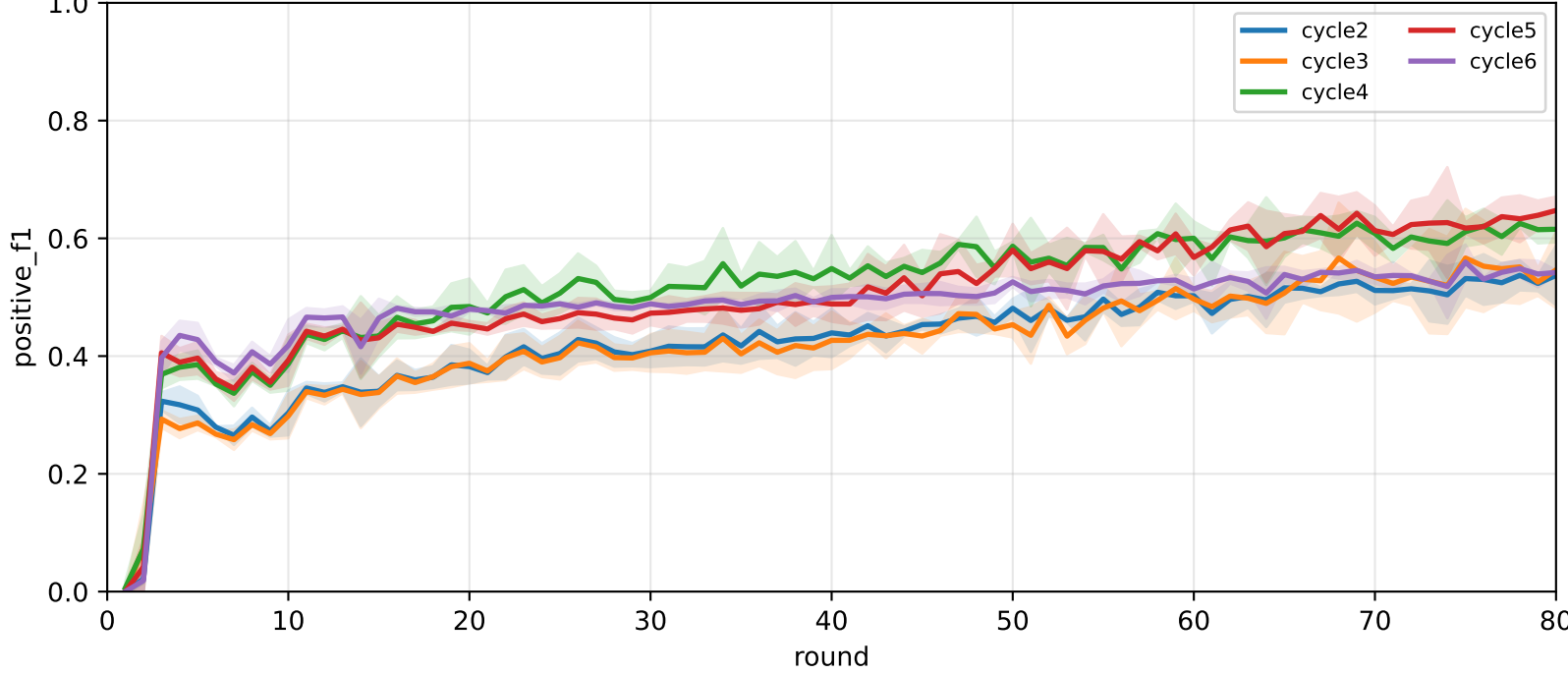
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



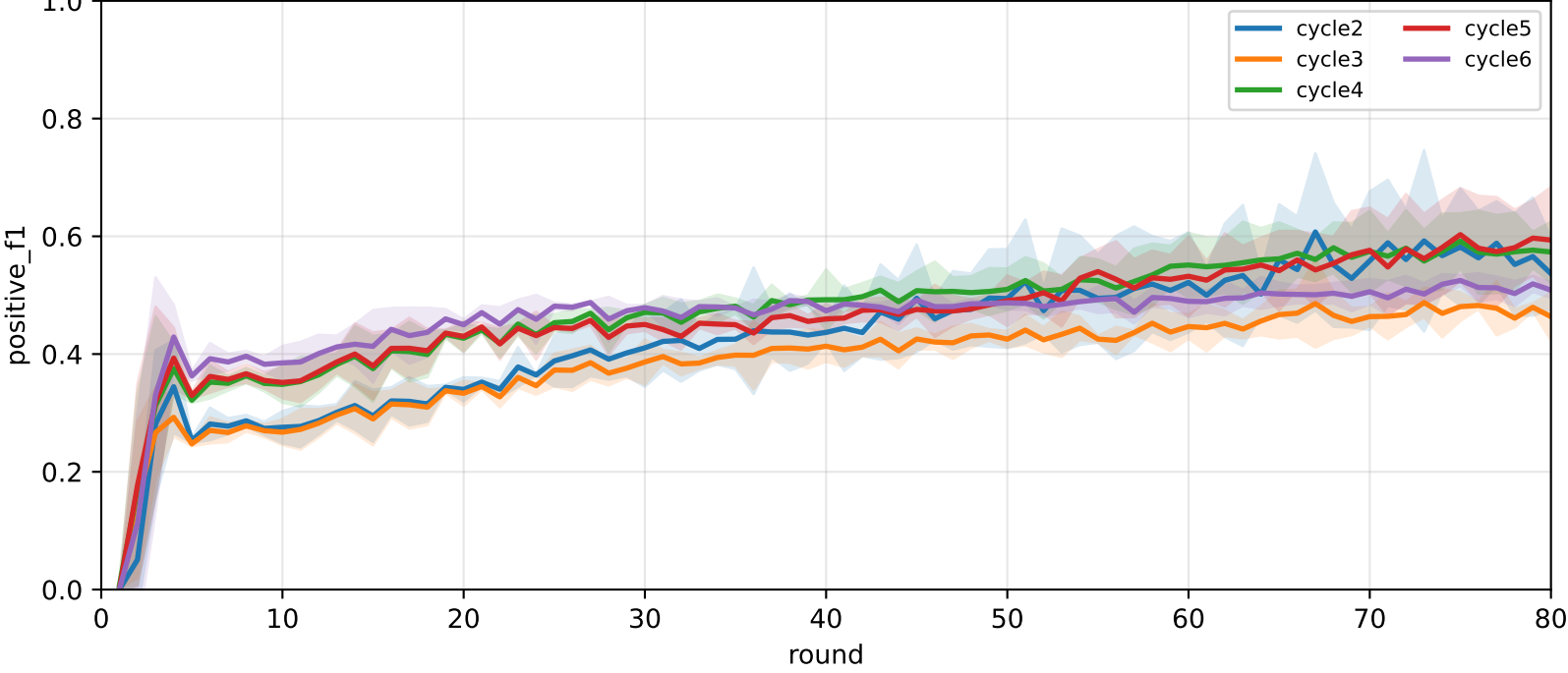
Pooled local centralized positive F1



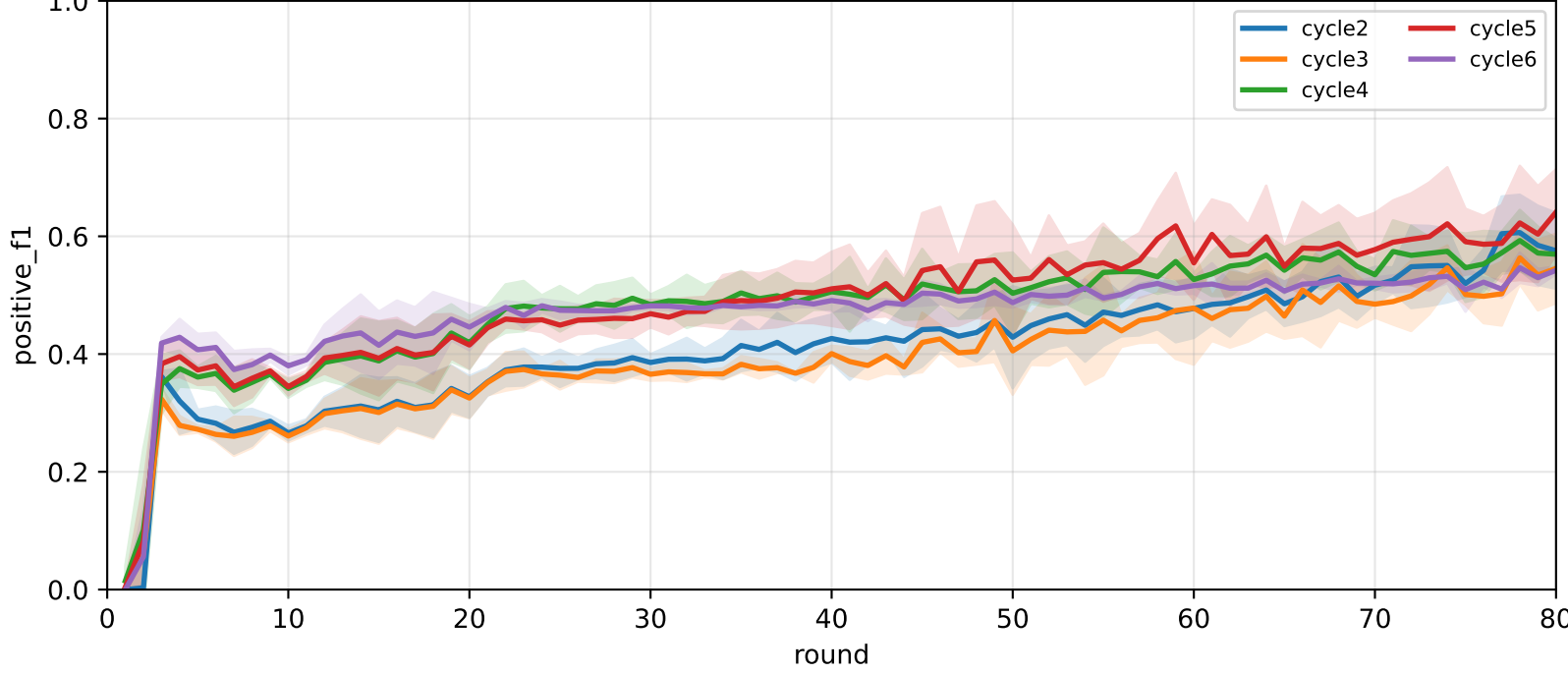
Pooled FedAvg positive F1



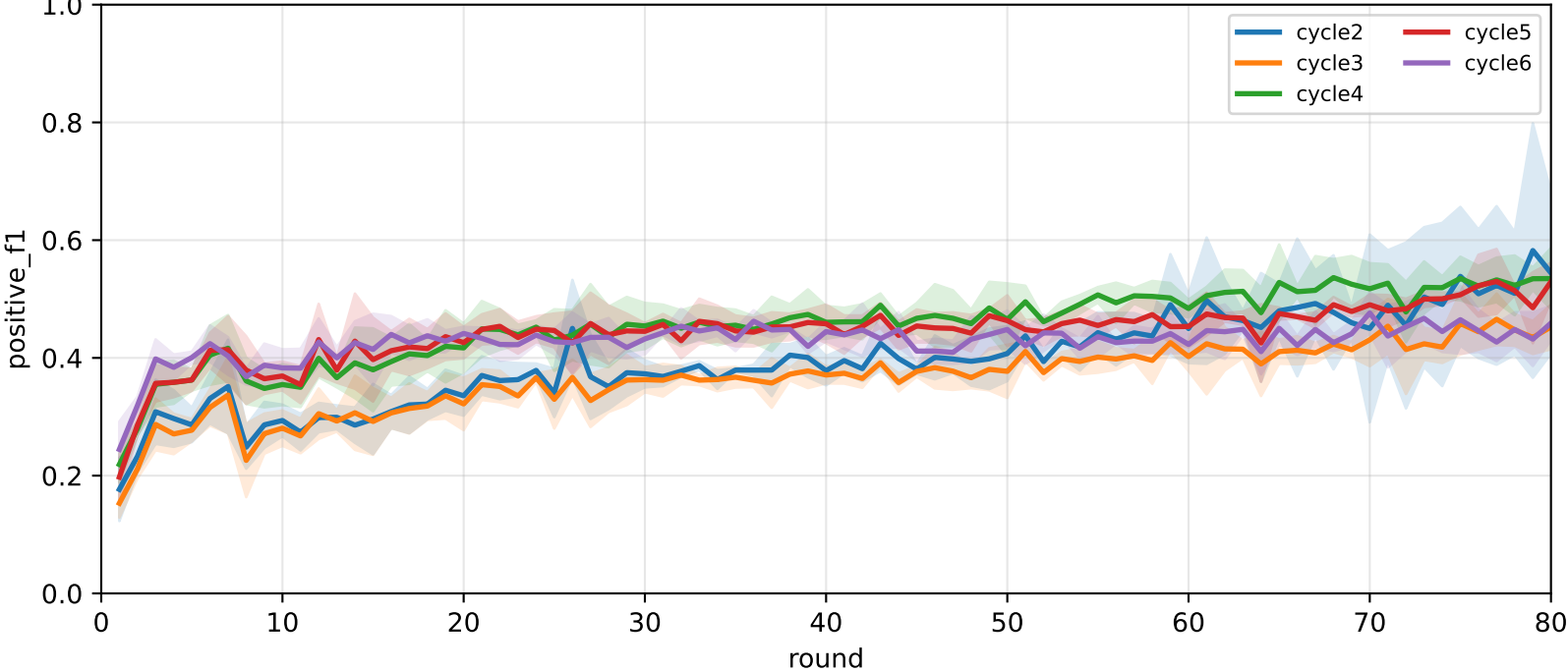
Pooled FedProx positive F1



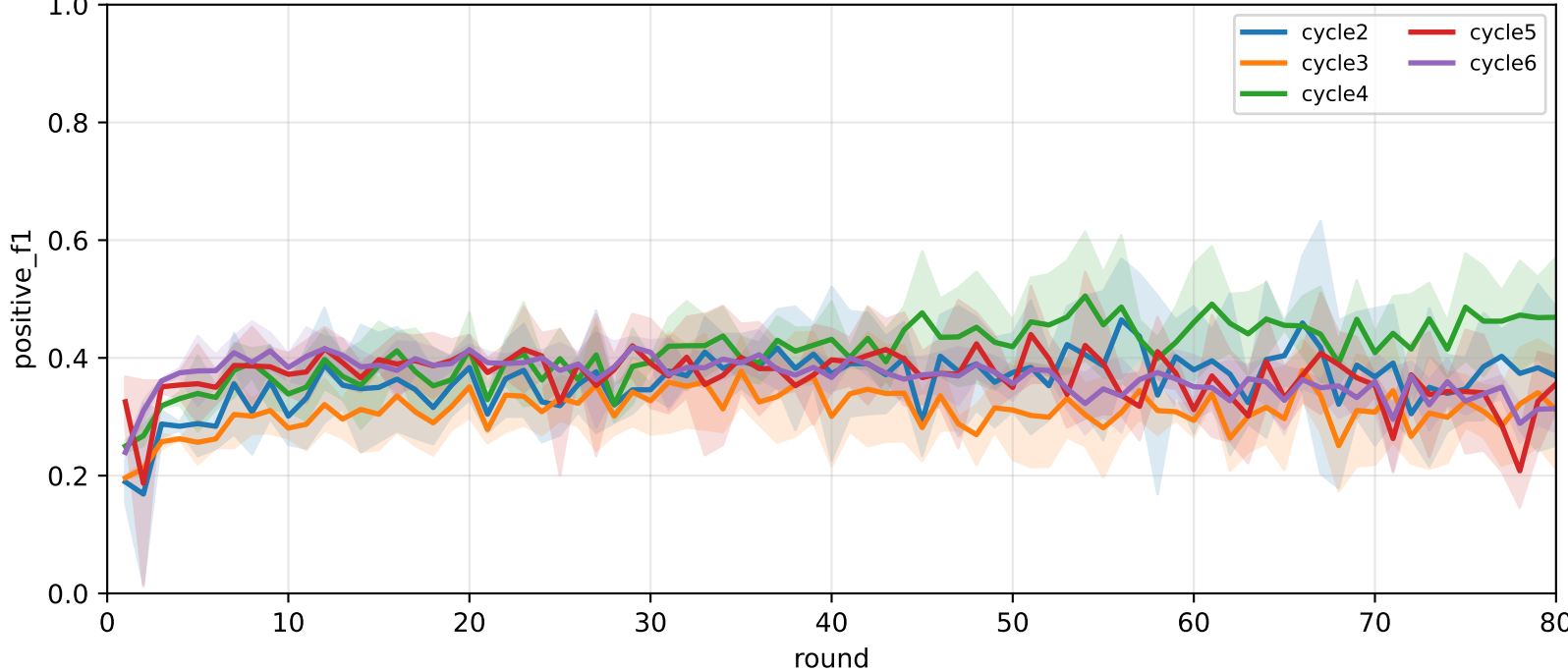
Pooled GCFL+ positive F1



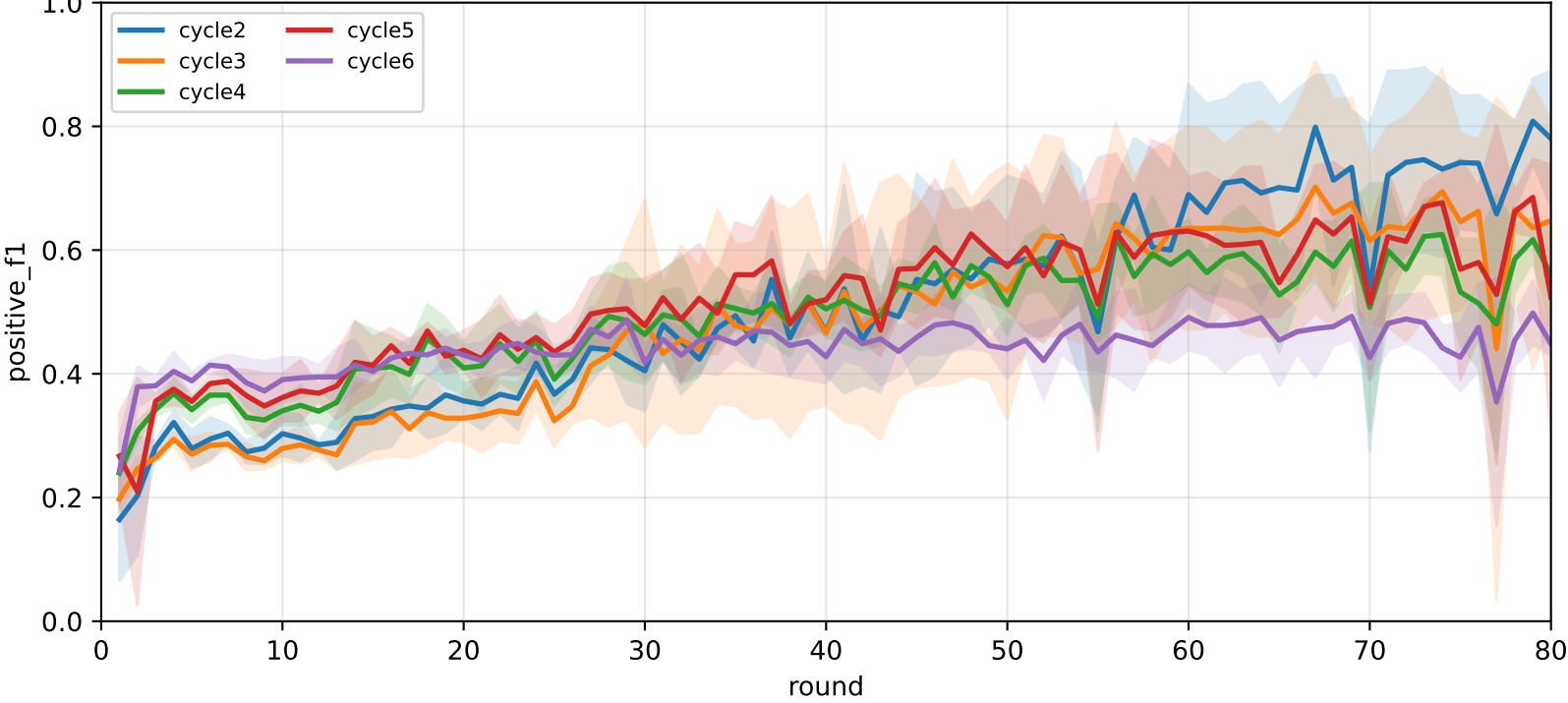
Pooled APPLE positive F1 (after-local selected model, rounds=80)



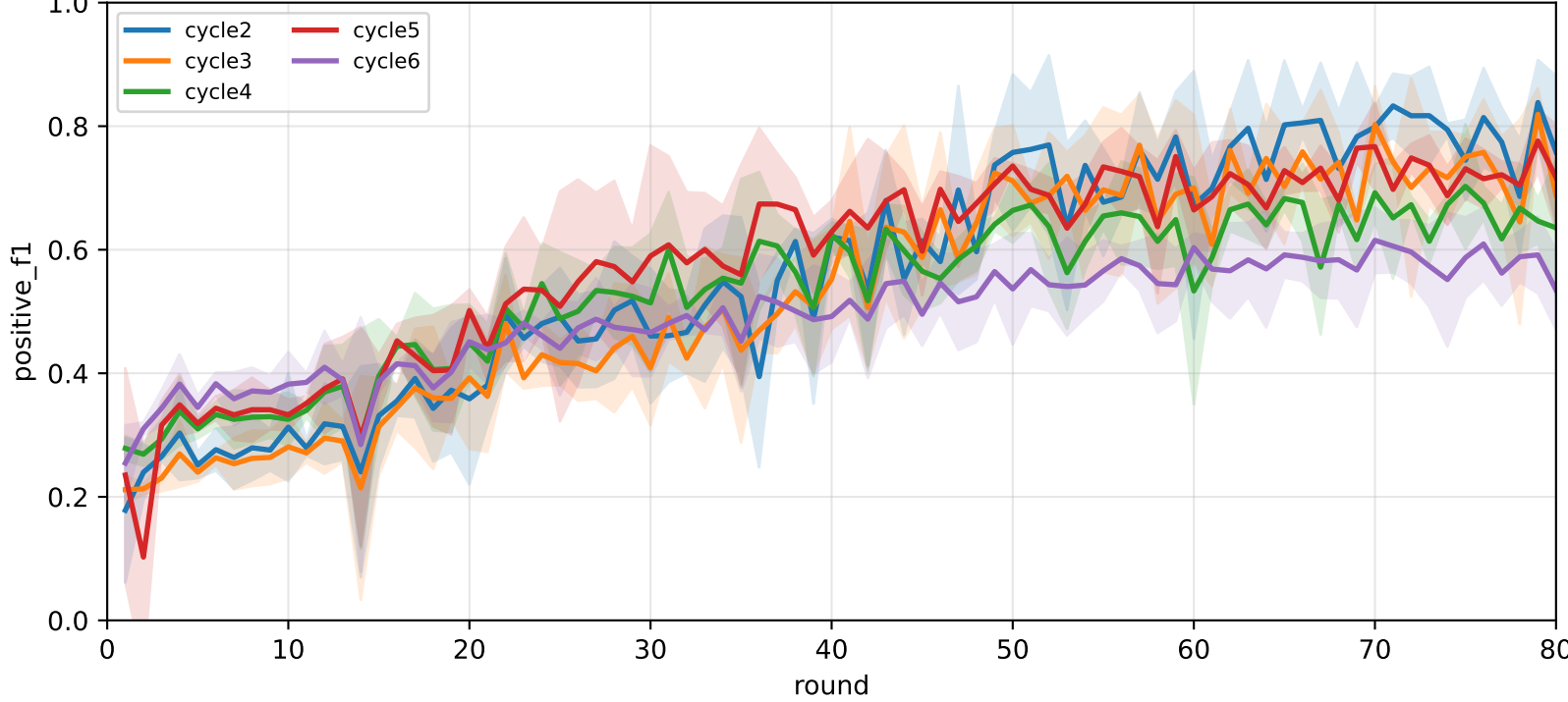
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



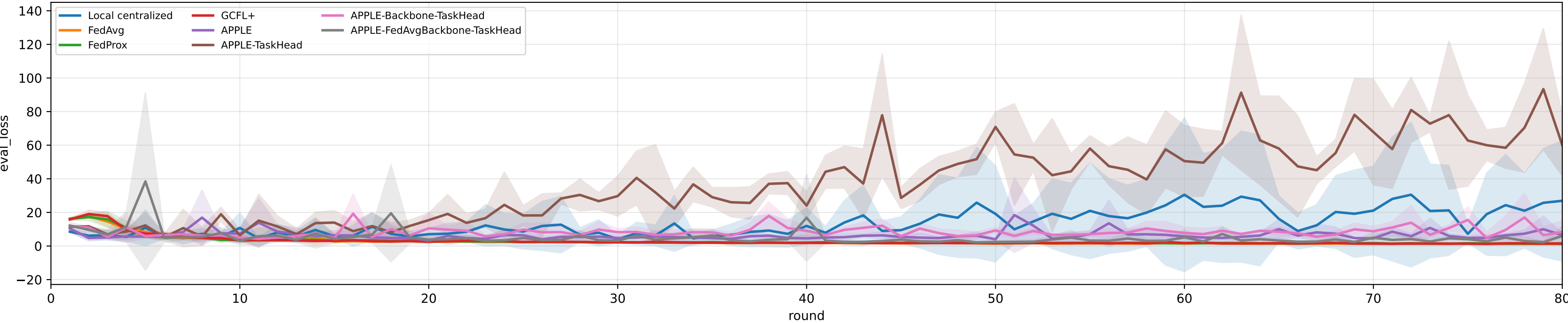
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

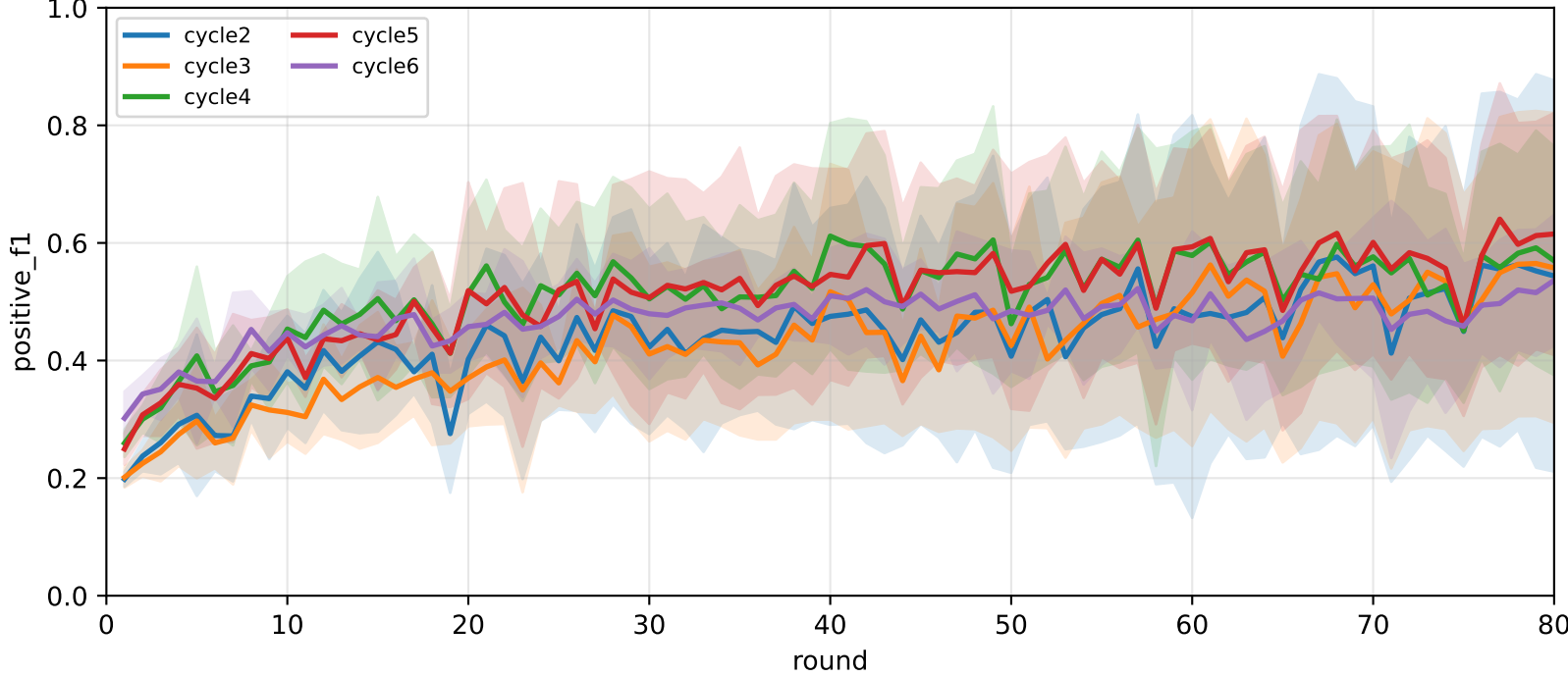
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	34.0% ± 13.0%	34.3% ± 13.3%	29.8% ± 14.7%	32.8% ± 11.9%	34.7% ± 10.6%	37.8% ± 19.0%	36.2% ± 10.7%
FedAvg	54.2% ± 5.0%	54.7% ± 5.1%	50.1% ± 5.1%	60.4% ± 7.4%	56.9% ± 5.7%	56.6% ± 5.7%	49.4% ± 4.5%
FedProx	52.5% ± 2.7%	52.9% ± 2.7%	53.1% ± 3.0%	55.0% ± 3.4%	54.5% ± 5.6%	53.2% ± 5.8%	48.7% ± 1.0%
GCFL+	53.9% ± 1.4%	54.5% ± 1.6%	53.6% ± 4.8%	58.0% ± 4.2%	53.8% ± 4.2%	56.9% ± 5.7%	49.8% ± 1.0%
APPLE	27.1% ± 3.9%	27.1% ± 3.8%	22.5% ± 3.6%	27.8% ± 4.4%	27.7% ± 4.1%	27.1% ± 3.6%	30.5% ± 3.6%
APPLE-TaskHead	32.1% ± 5.5%	32.1% ± 5.5%	27.3% ± 6.3%	32.2% ± 6.3%	33.3% ± 5.8%	31.6% ± 5.7%	36.0% ± 3.8%
APPLE-Backbone-TaskHead	40.3% ± 7.6%	40.2% ± 7.6%	34.8% ± 7.6%	43.0% ± 9.4%	41.7% ± 8.2%	40.8% ± 9.2%	40.9% ± 4.6%
APPLE-FedAvgBackbone-TaskHead	56.7% ± 3.8%	57.7% ± 3.9%	59.9% ± 9.0%	61.3% ± 5.3%	58.8% ± 3.7%	59.1% ± 6.8%	49.4% ± 3.6%

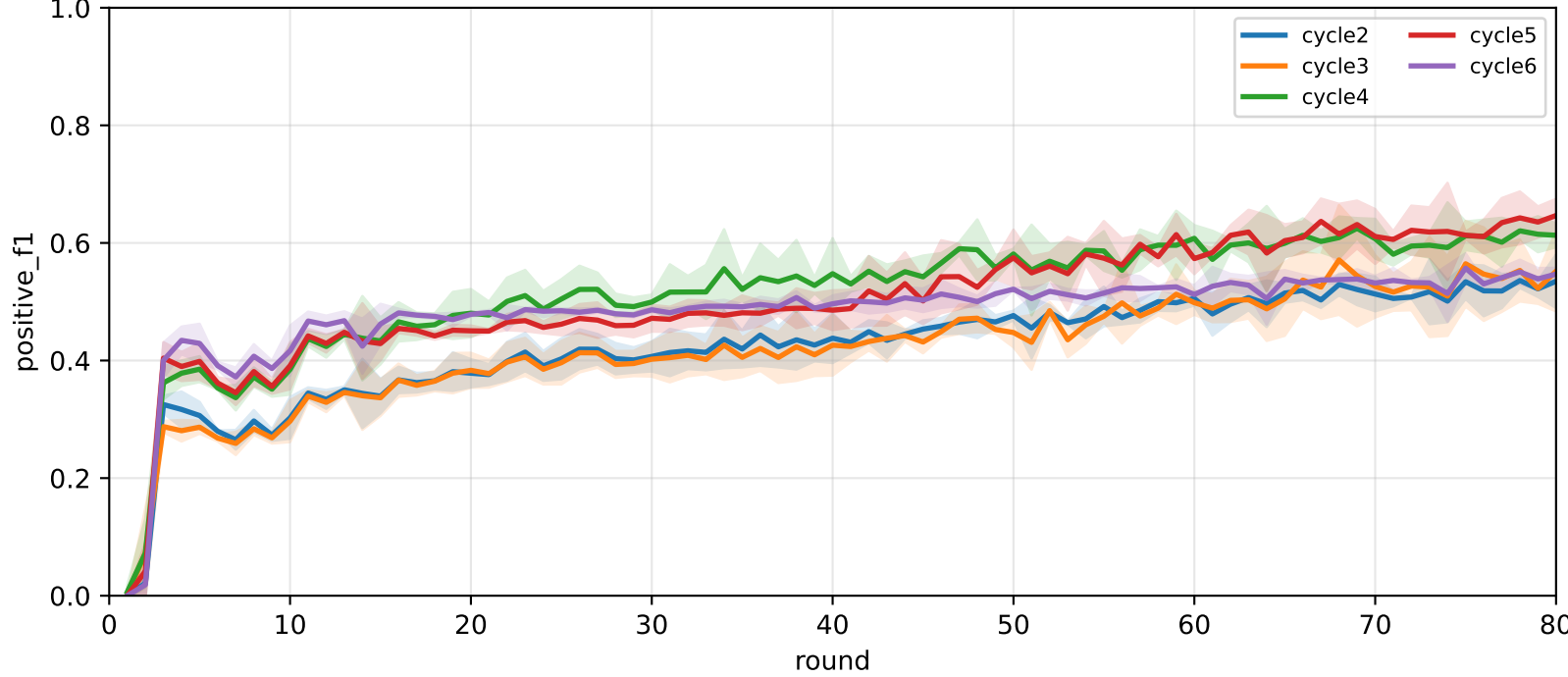
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



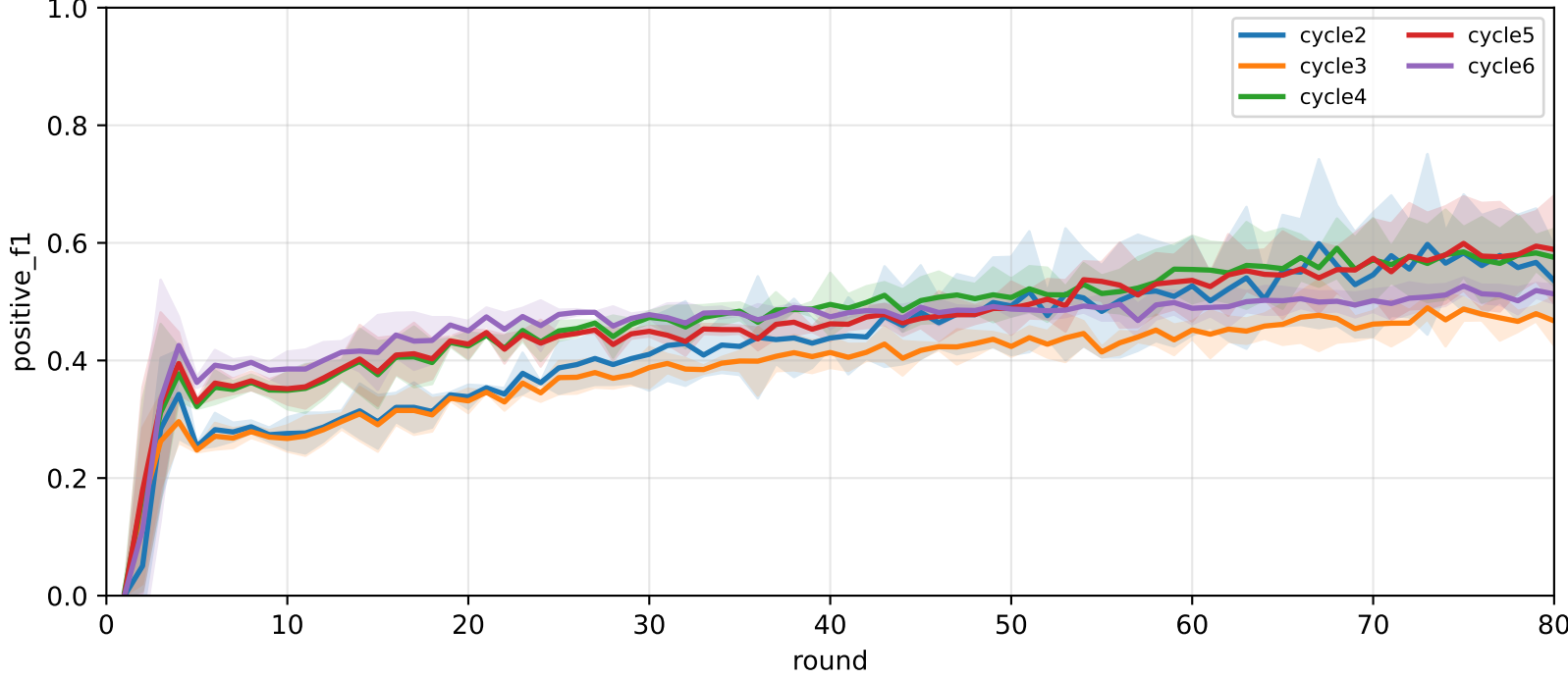
Pooled local centralized positive F1



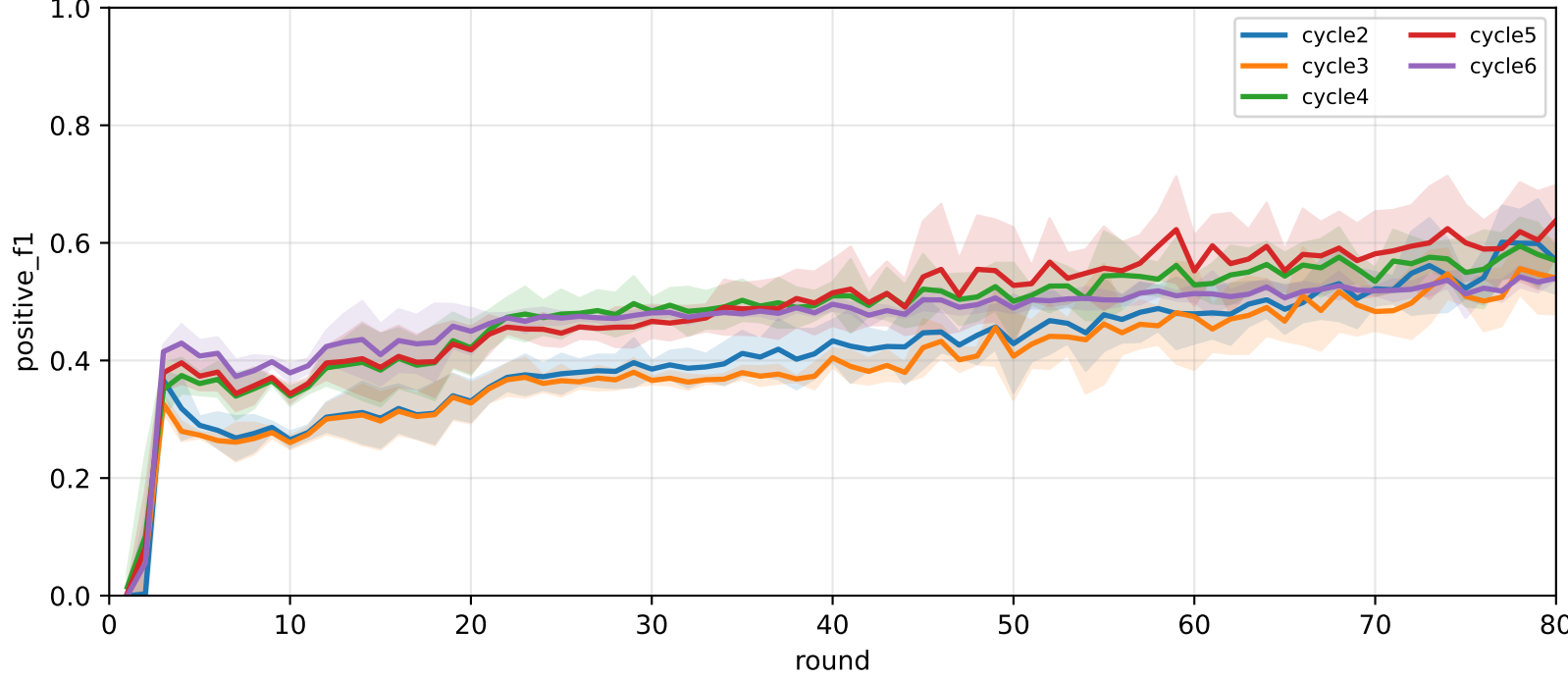
Pooled FedAvg positive F1



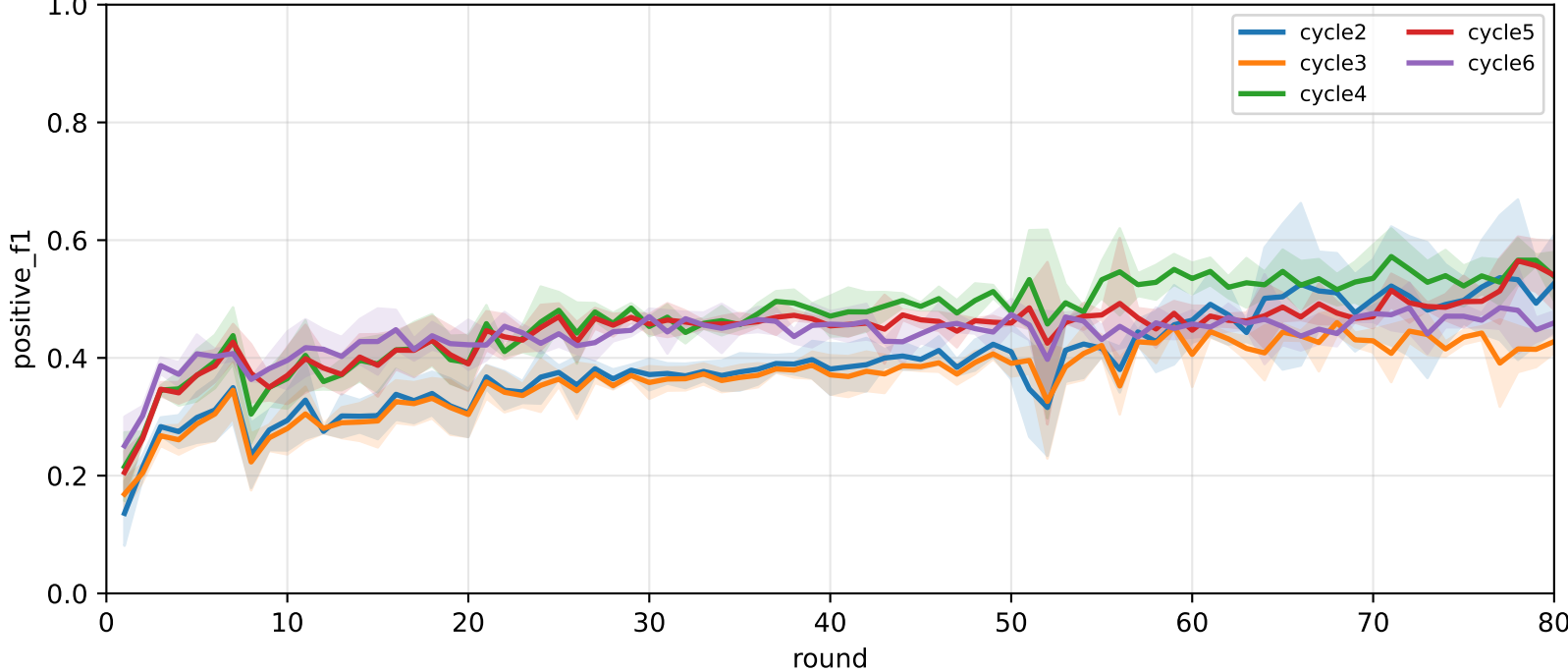
Pooled FedProx positive F1



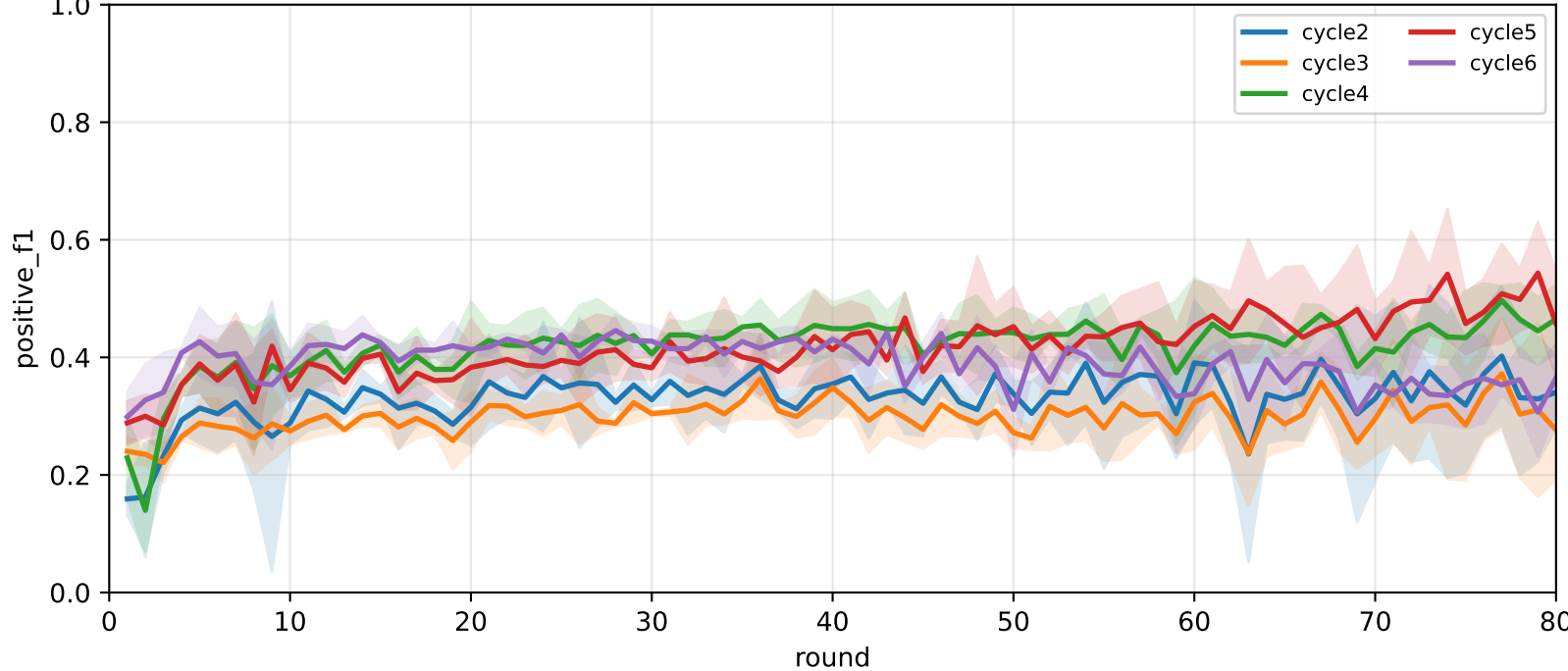
Pooled GCFL+ positive F1



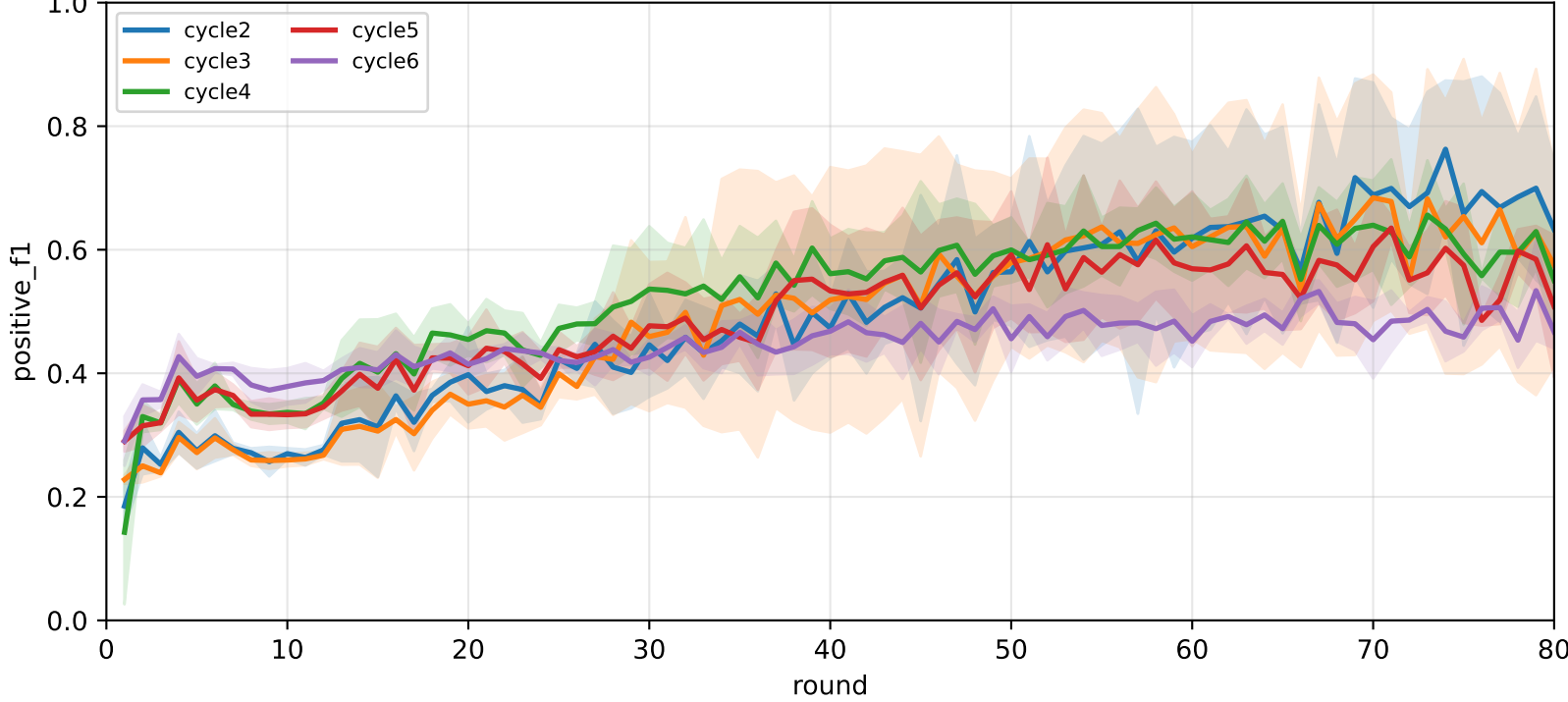
Pooled APPLE positive F1 (after-local selected model, rounds=80)



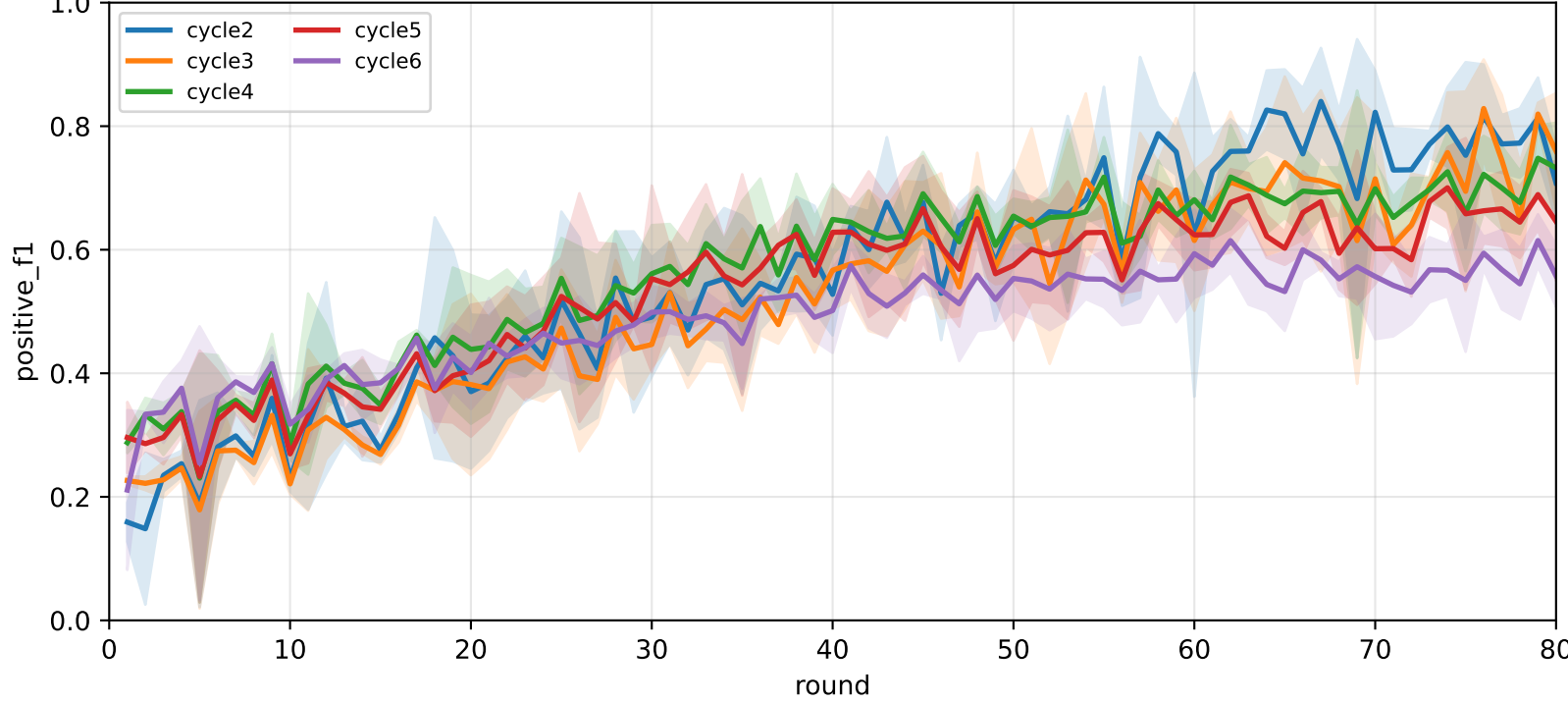
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



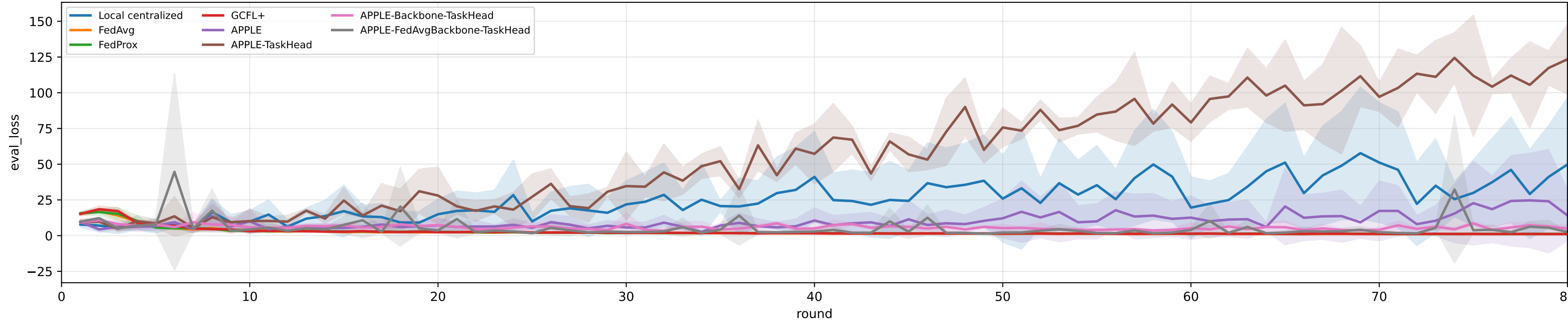
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

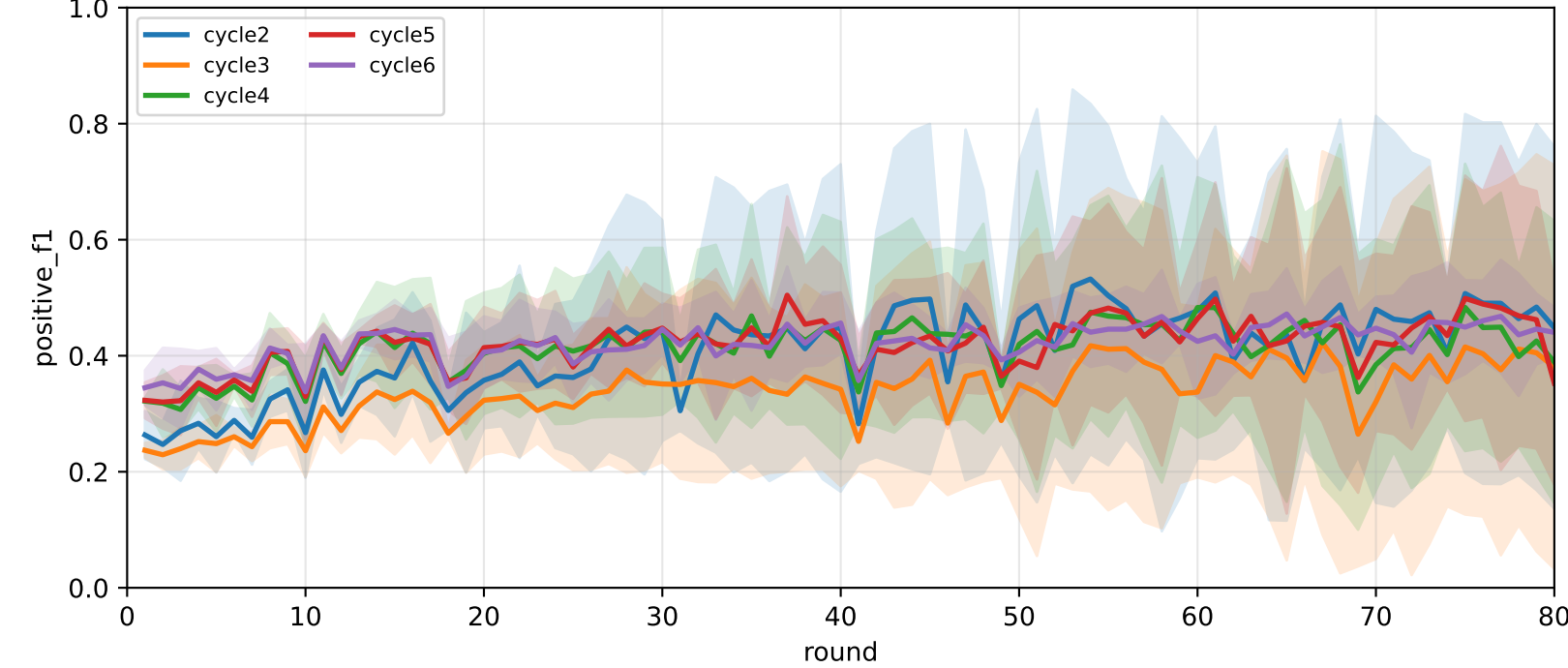
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	33.5% ± 12.2%	34.3% ± 13.3%	36.4% ± 23.0%	36.9% ± 15.6%	32.4% ± 10.8%	32.2% ± 9.6%	33.5% ± 7.5%
FedAvg	54.6% ± 4.8%	55.1% ± 4.9%	50.3% ± 4.8%	60.8% ± 7.1%	56.7% ± 5.2%	58.0% ± 6.0%	49.6% ± 4.4%
FedProx	52.7% ± 3.3%	53.1% ± 3.3%	53.5% ± 2.4%	55.1% ± 4.0%	54.5% ± 6.3%	53.6% ± 6.4%	48.9% ± 1.5%
GCFL+	53.9% ± 0.7%	54.5% ± 0.8%	53.4% ± 4.5%	58.0% ± 3.2%	53.9% ± 4.6%	57.0% ± 5.3%	49.9% ± 1.1%
APPLE	28.0% ± 1.4%	28.0% ± 1.4%	23.1% ± 1.2%	28.6% ± 1.9%	29.1% ± 0.9%	27.5% ± 1.4%	31.5% ± 1.4%
APPLE-TaskHead	30.3% ± 2.1%	30.3% ± 2.1%	25.2% ± 2.7%	31.1% ± 2.8%	30.8% ± 2.2%	31.7% ± 2.1%	32.5% ± 1.4%
APPLE-Backbone-TaskHead	39.5% ± 3.9%	39.5% ± 4.1%	34.1% ± 5.7%	41.9% ± 6.7%	41.3% ± 3.6%	39.7% ± 3.1%	40.4% ± 2.9%
APPLE-FedAvgBackbone-TaskHead	60.3% ± 2.8%	62.3% ± 3.3%	64.5% ± 8.1%	67.2% ± 8.5%	67.1% ± 3.8%	63.8% ± 2.5%	48.8% ± 4.1%

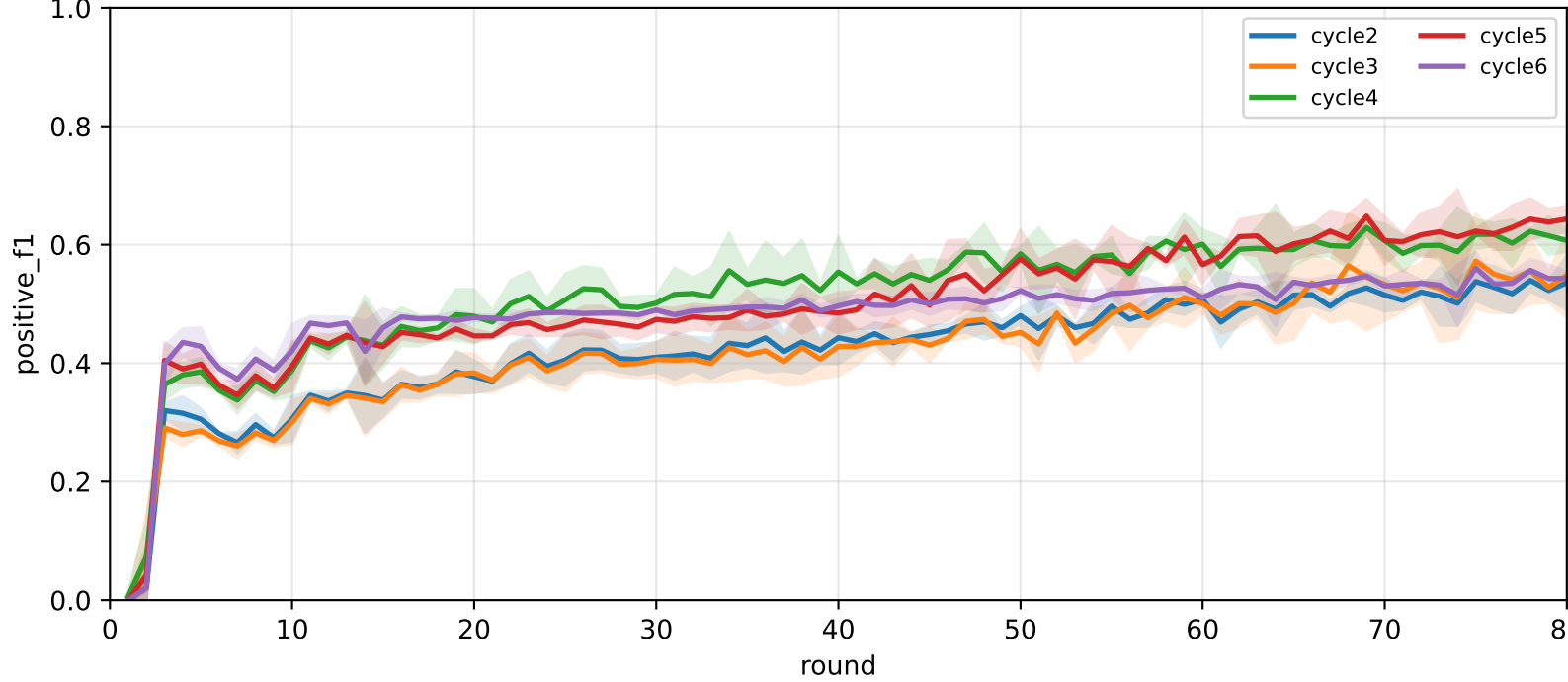
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



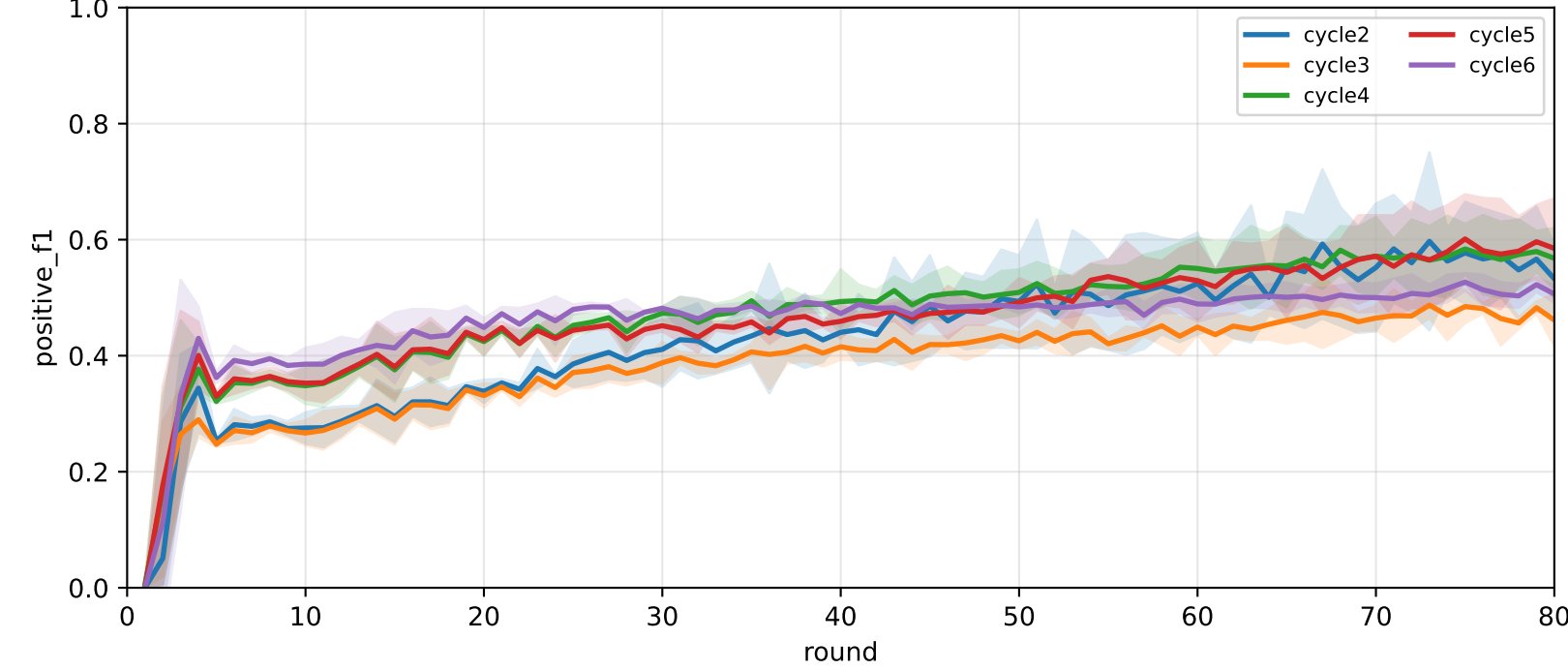
Pooled local centralized positive F1



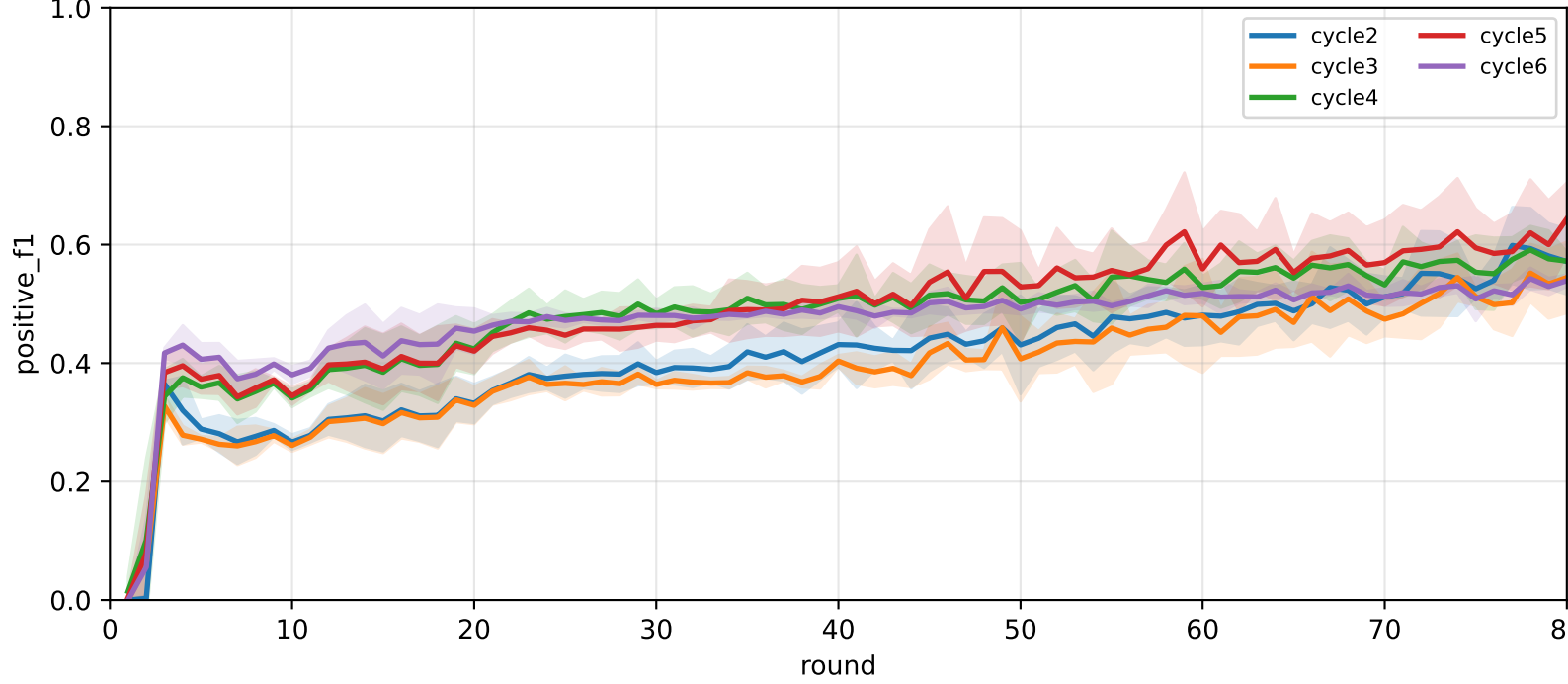
Pooled FedAvg positive F1



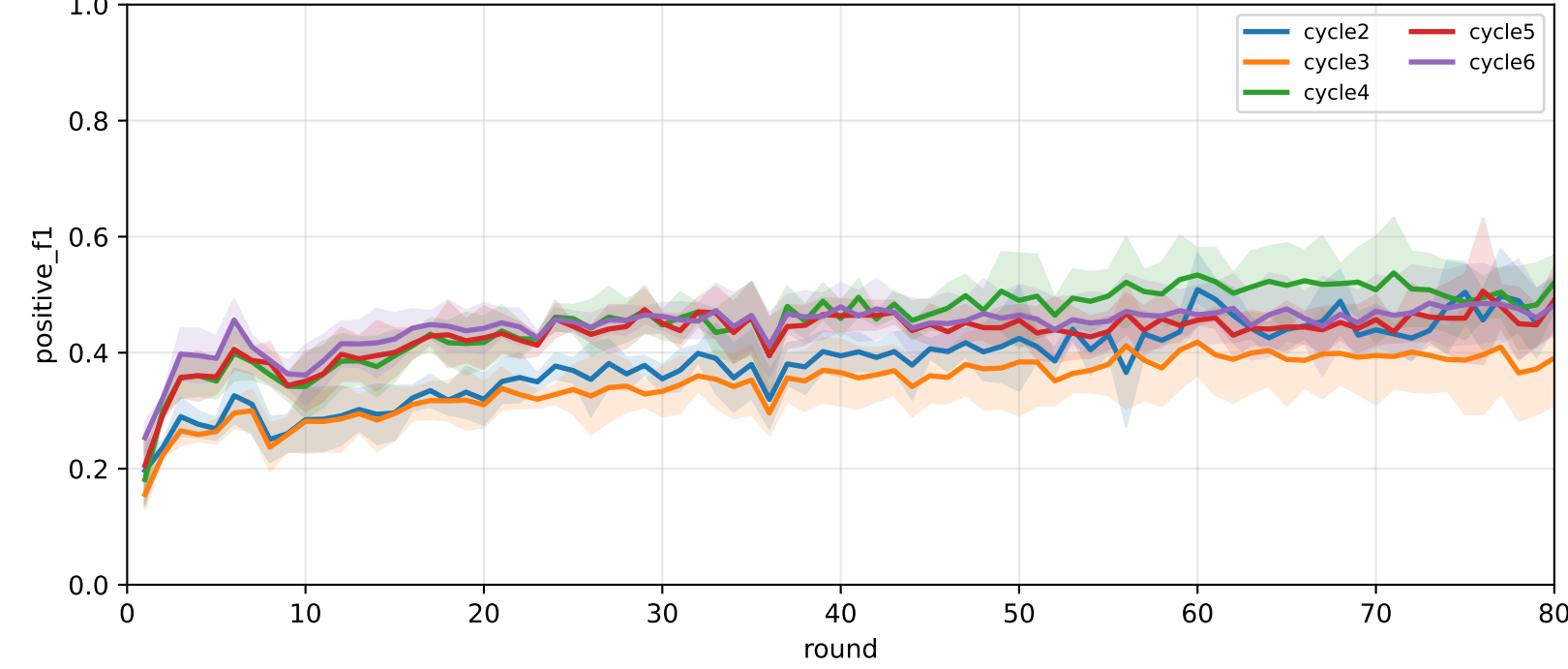
Pooled FedProx positive F1



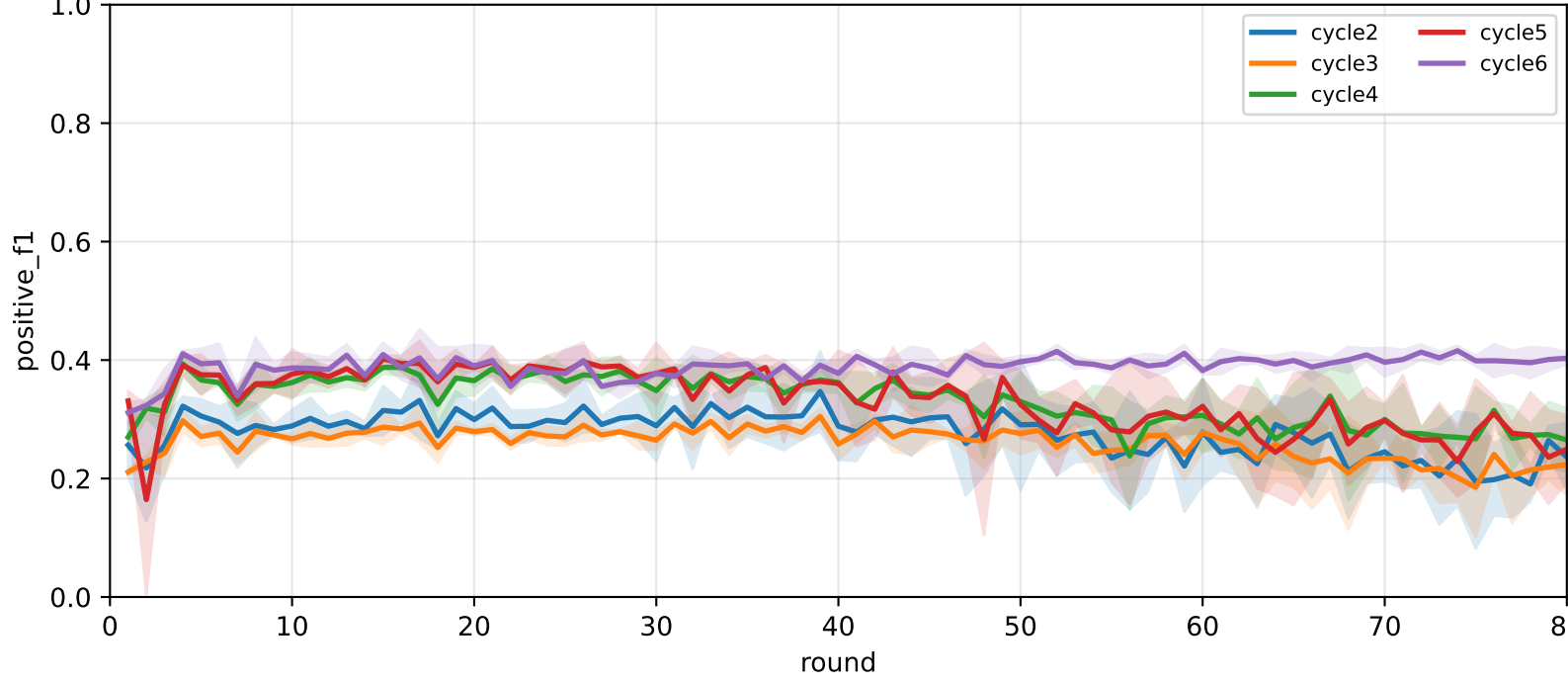
Pooled GCFL+ positive F1



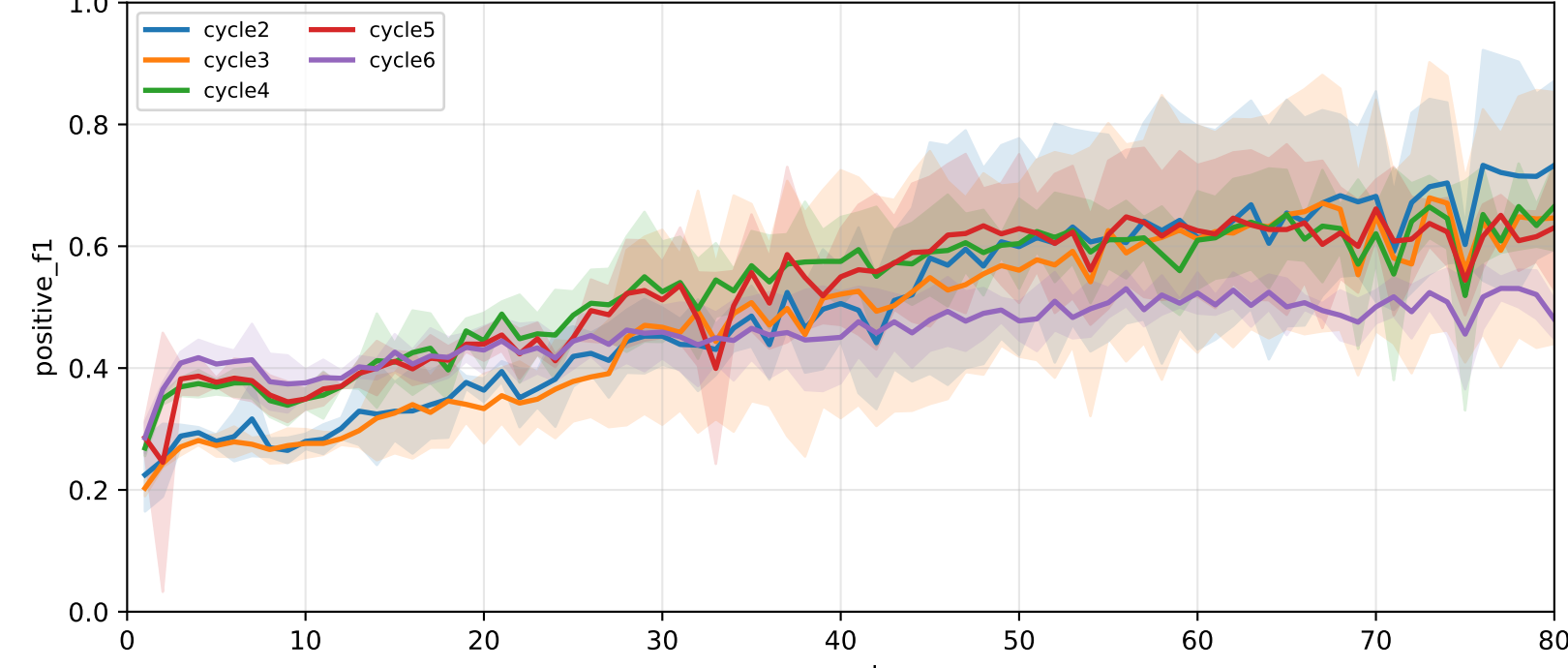
Pooled APPLE positive F1 (after-local selected model, rounds=80)



Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

